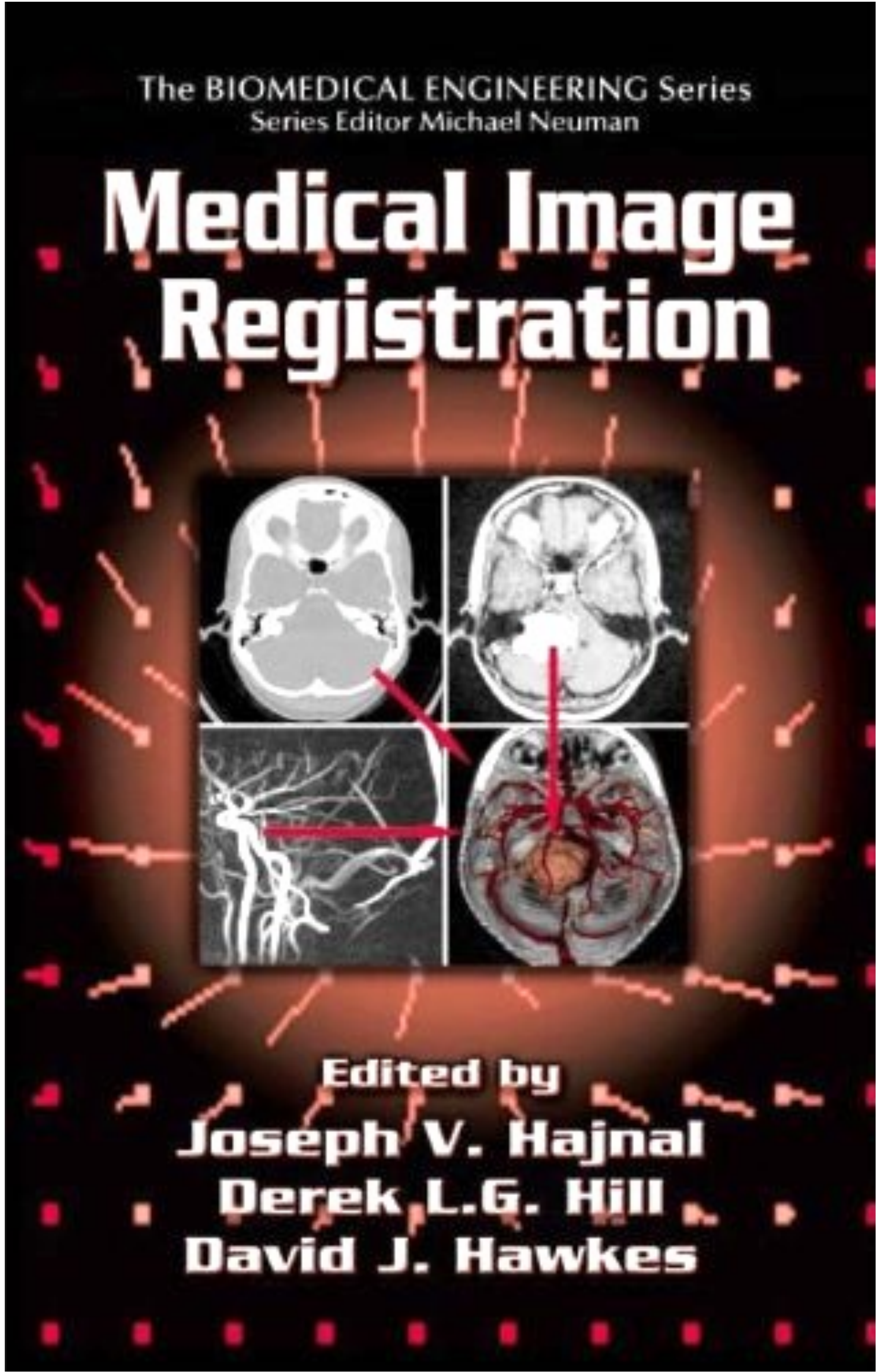
CS 736
Medical Image Computing (MIC)

Image Registration

Suyash P. Awate

References


The BIOMEDICAL ENGINEERING Series
Series Editor Michael Neuman
Medical Image Registration
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Joseph V. Hajnal
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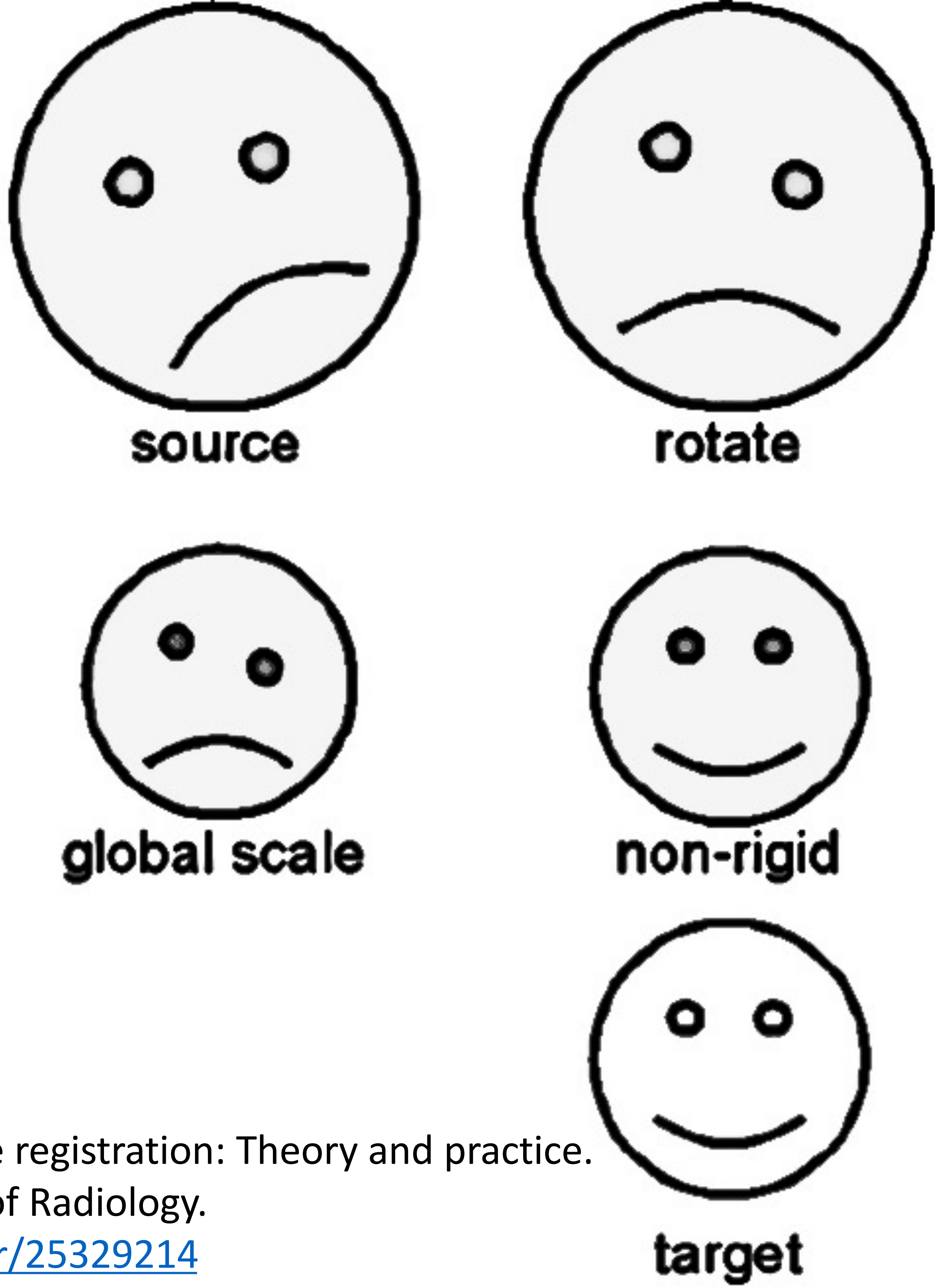
Image Registration

- Given: Two images
- Goal: “Transform” one image to “align” itself to other
- Components

1) Model for transformation, i.e., spatial deformation/warp

- Warp one image to superimpose objects within image onto corresponding objects in another image
- Geometric transformation on spatial domain



Non-rigid image registration: Theory and practice. British Journal of Radiology. DOI:10.1259/bjr/25329214

Image Registration

- Image deformation/warping (distortion)
    - www.youtube.com/watch?v=JsR5PKmm7S0

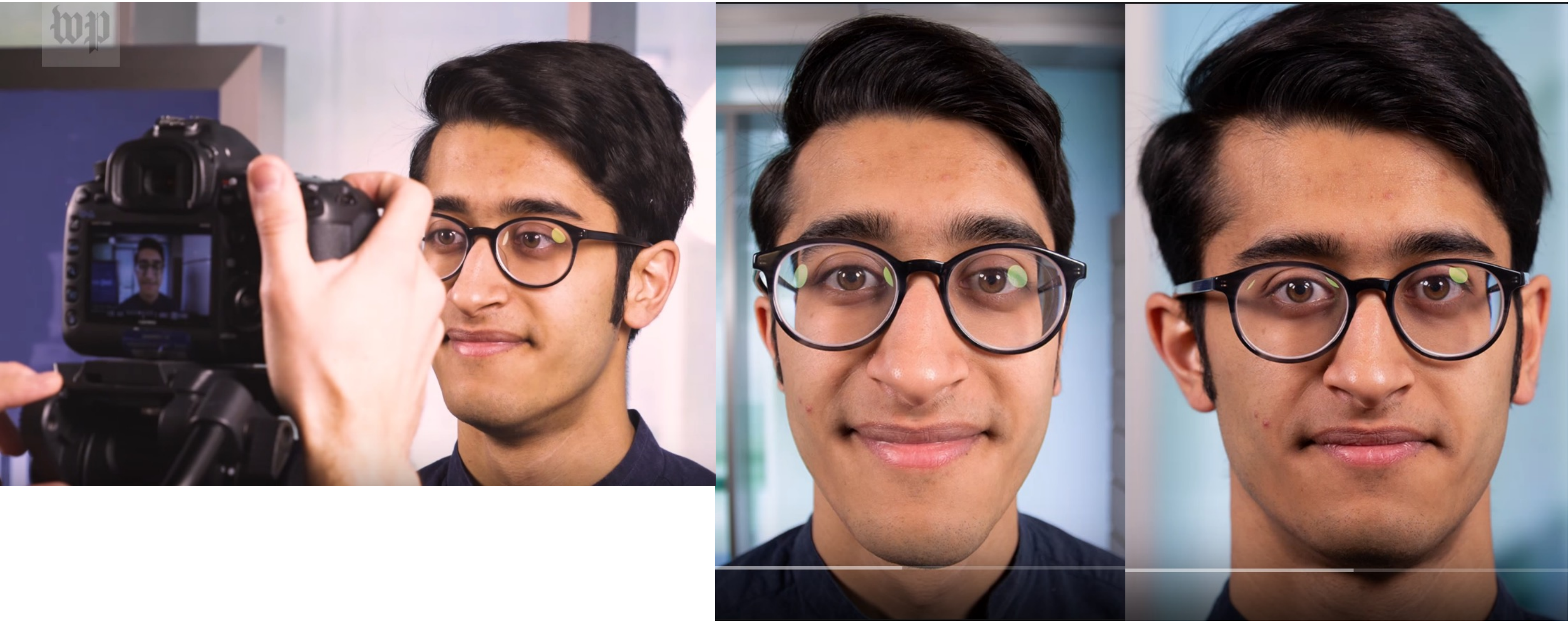


Image Registration

- Given: Two images
- Goal: “Transform” one image to “align” itself to other
- Components

1) Transformation model, i.e., spatial deformation / warp

- Warp one image to
  superimpose objects within image
  onto corresponding objects in another image
- Geometric transformation on spatial domain

2) Similarity measure between images

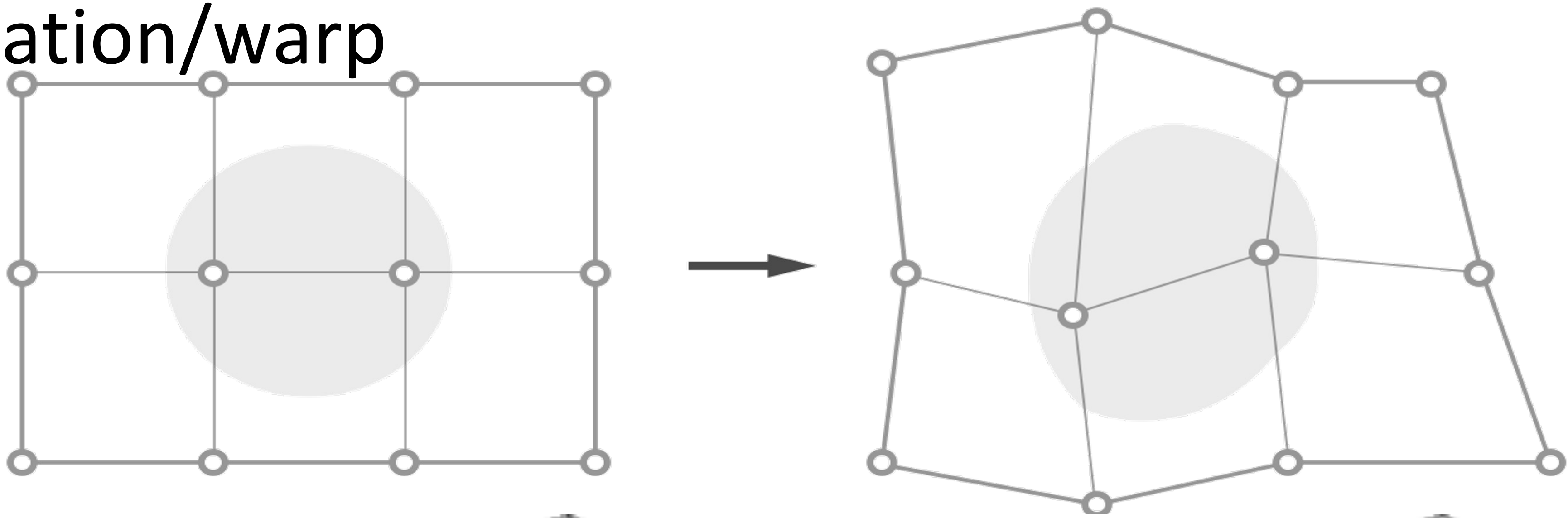
- To quantify how well
  warp superimposes corresponding objects between two images

Image Registration

- Components

1) Model for spatial deformation/warp

- Geometric transformation on spatial domain



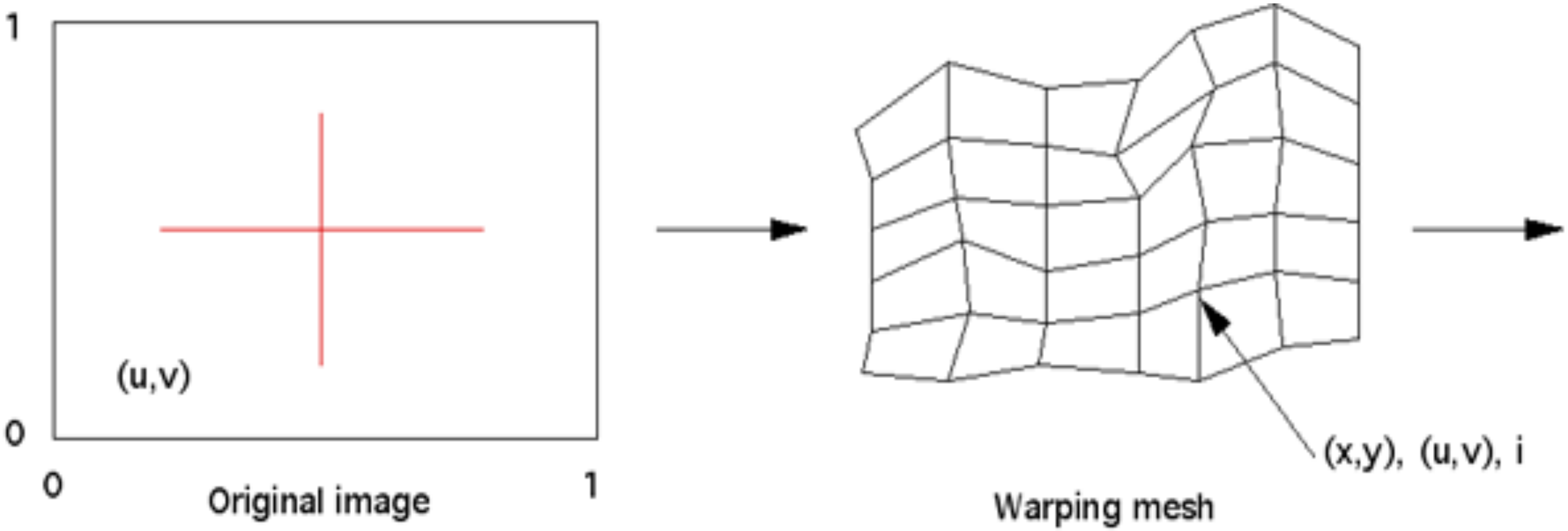
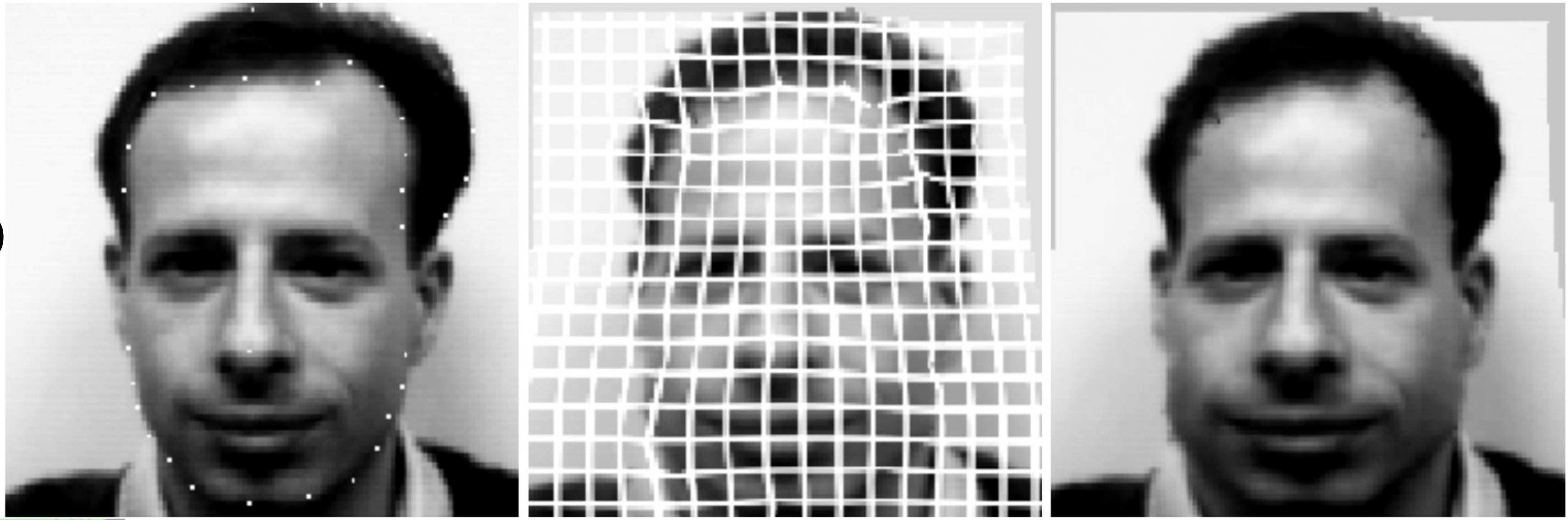


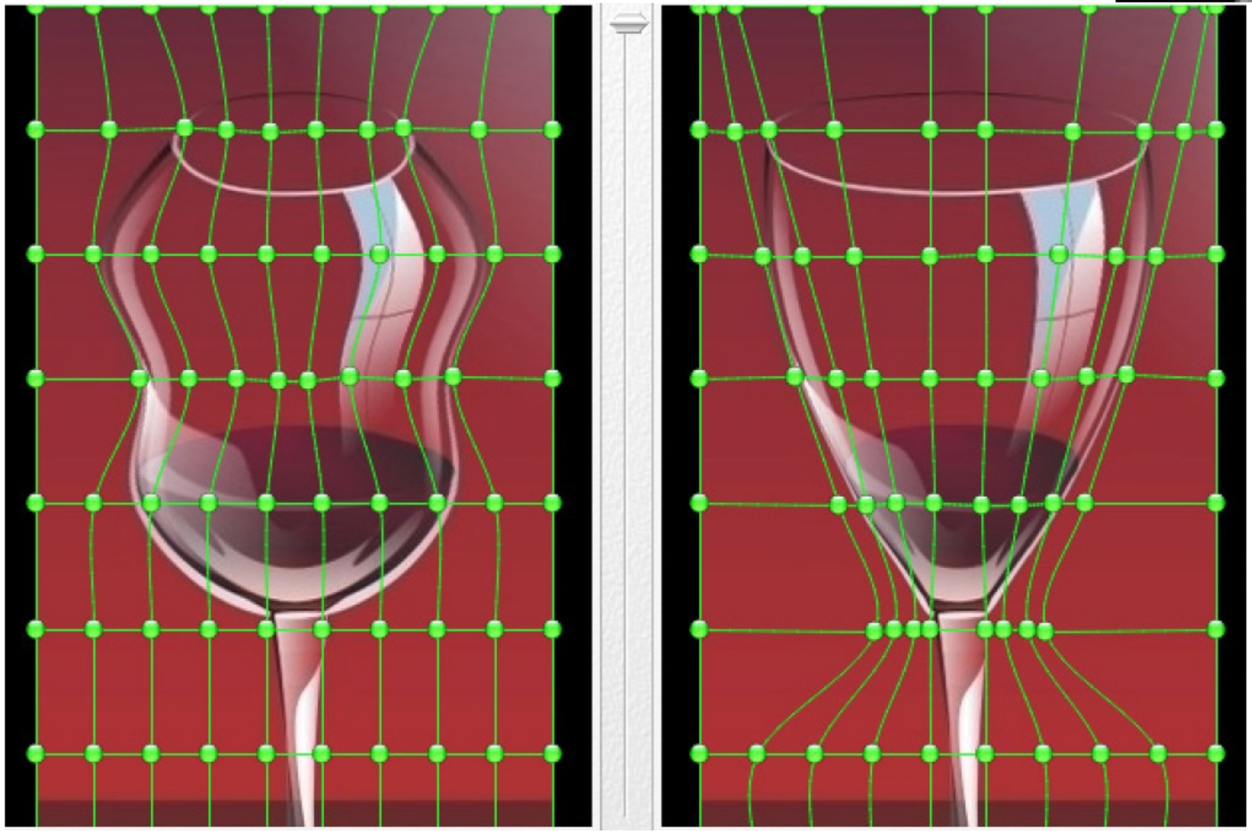
Image Registration

- Components

1) Model for
   spatial deformation/warp

- Geometric transformation
  on spatial domain





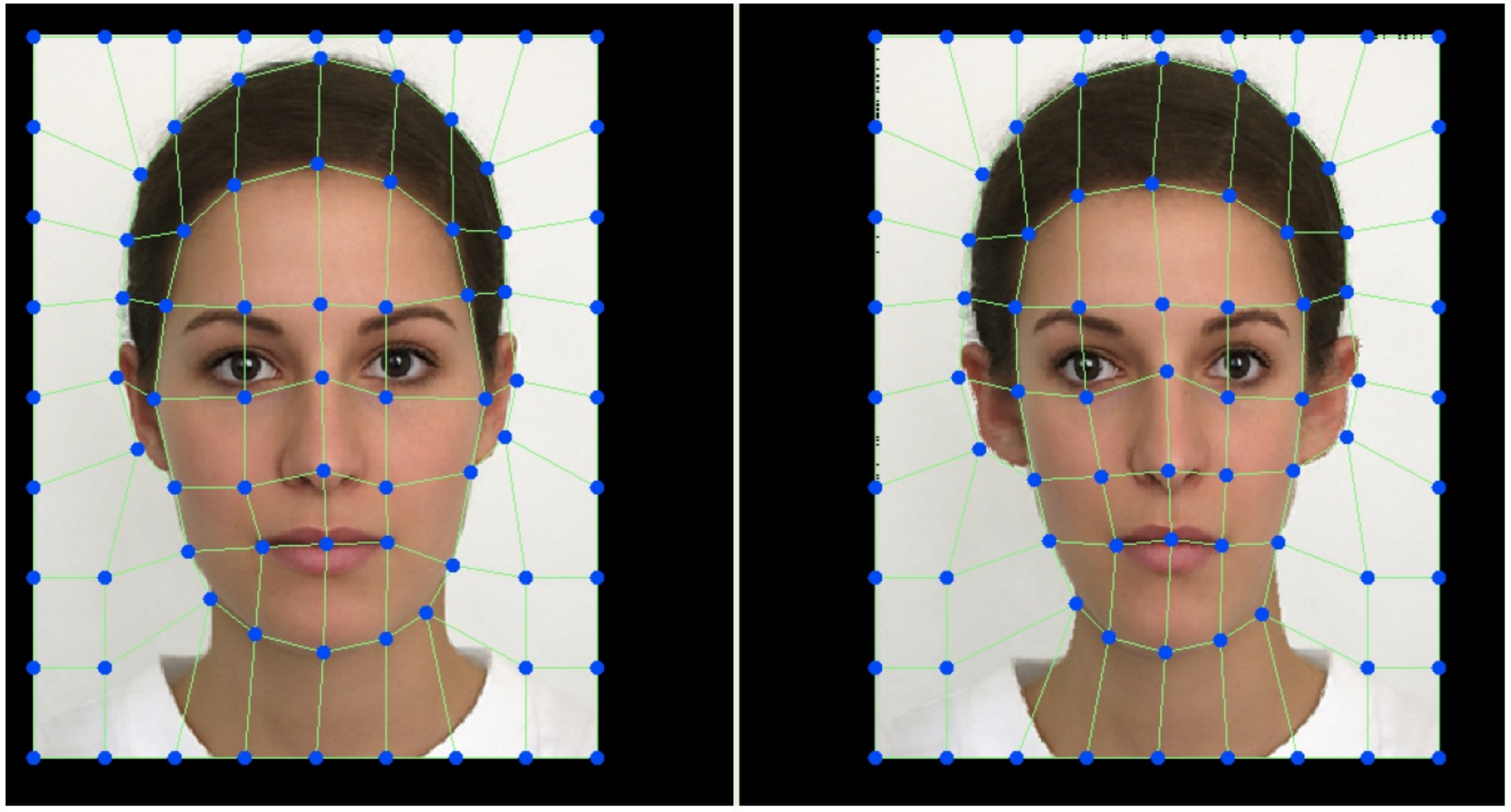
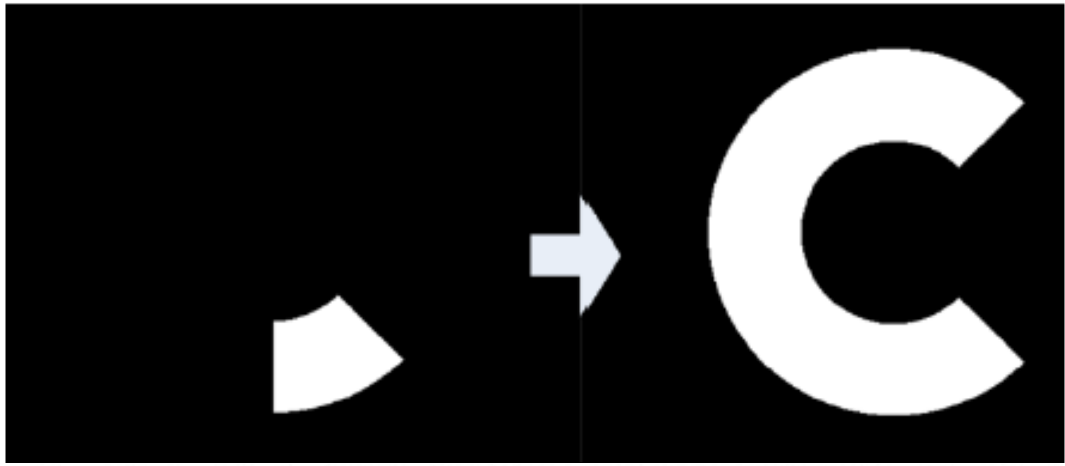


Image Registration



- Components

1) Model for spatial deformation/warp

- Geometric transformation on spatial domain

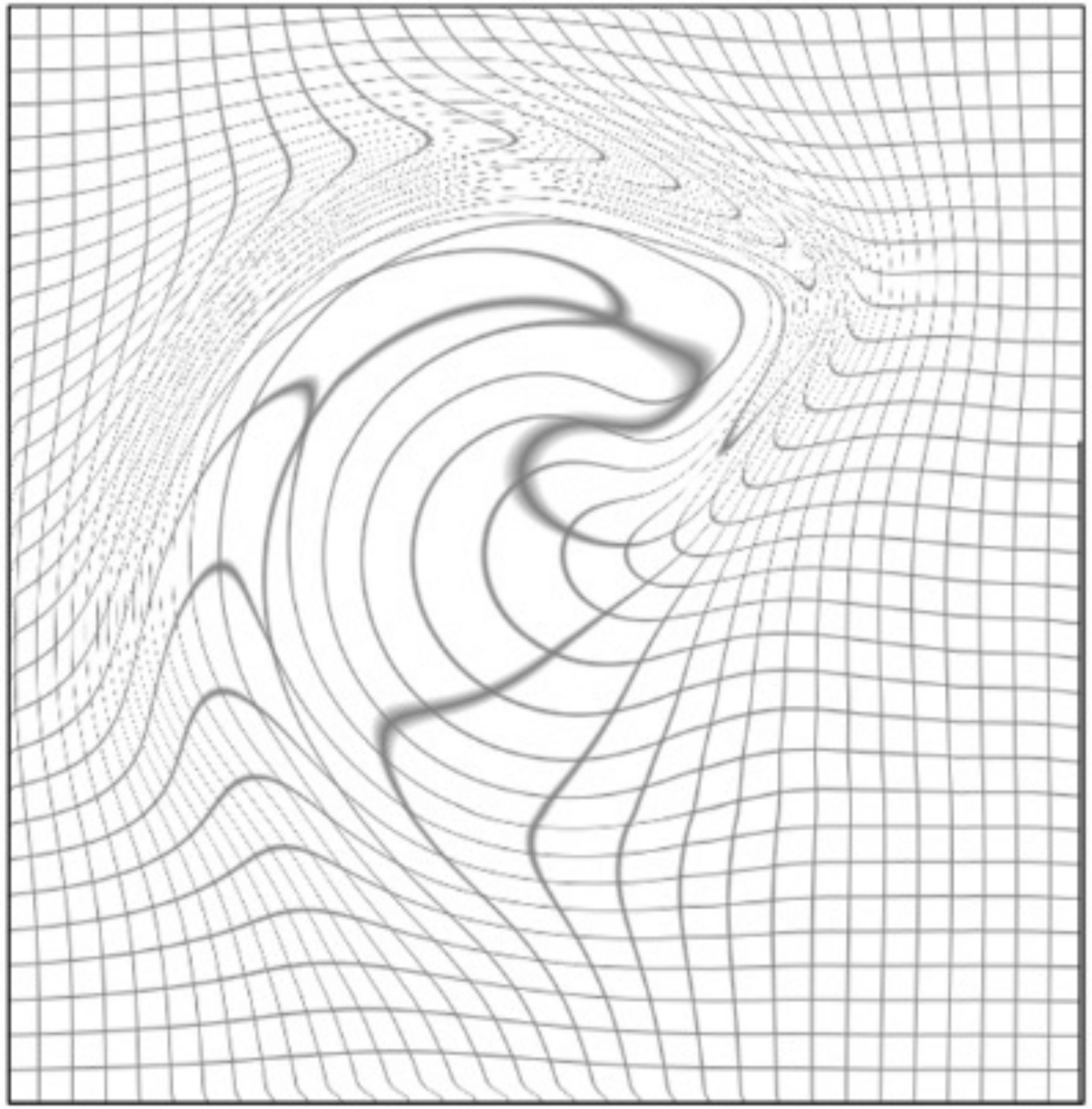


Image Registration

- Components

1) Model for spatial deformation

- Rigid transformation
- Similarity transformation
- Linear transformation
- Nonlinear transformation
  - Diffeomorphic (smooth, invertible)

  a) Parametric, e.g., B spline
  b) Nonparametric/dense, e.g., vector field

Image Registration

2) Similarity measure between images

- If images of same modality (e.g., T1w MRI and T1w MRI) → can use squared error between corresponding voxel intensities

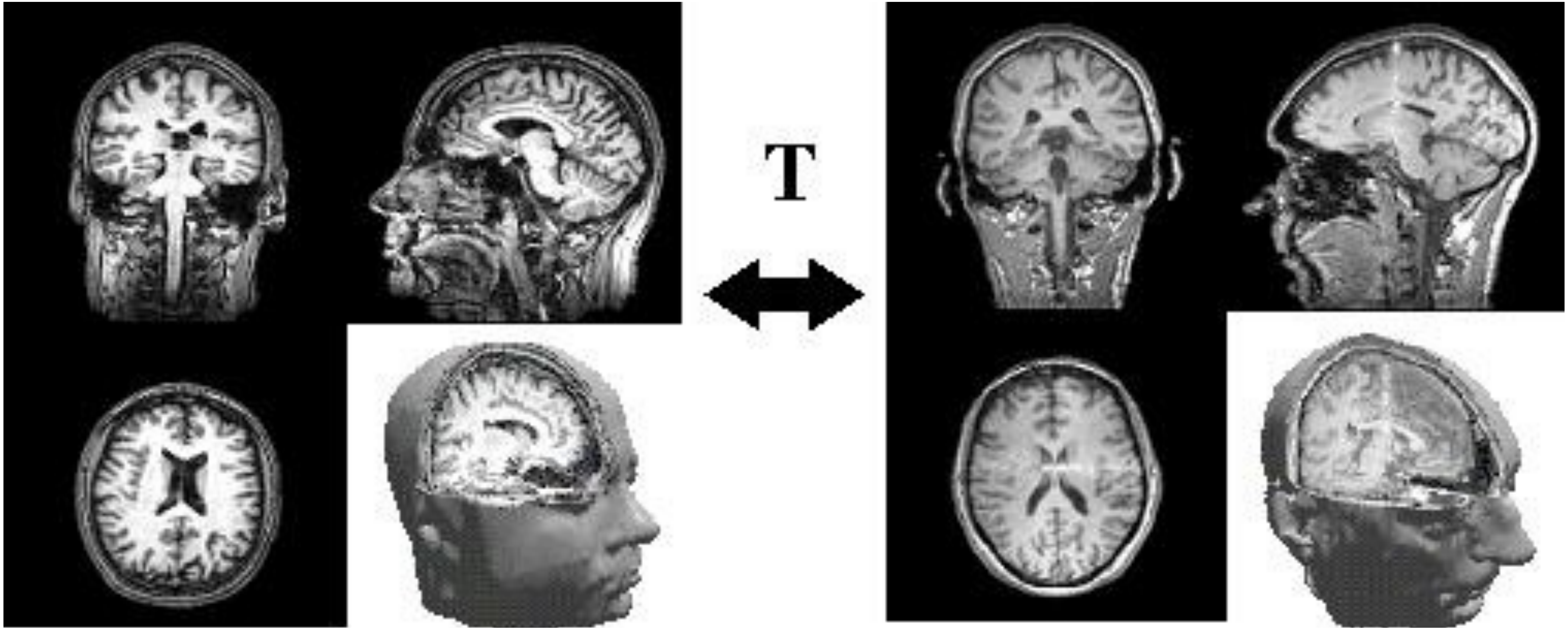
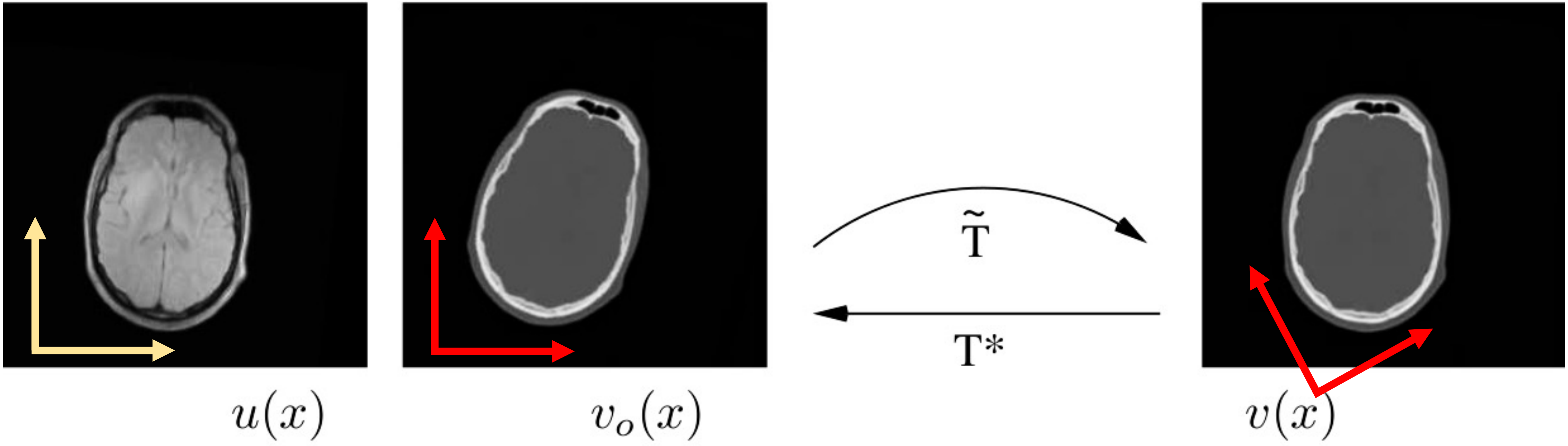


Image Registration

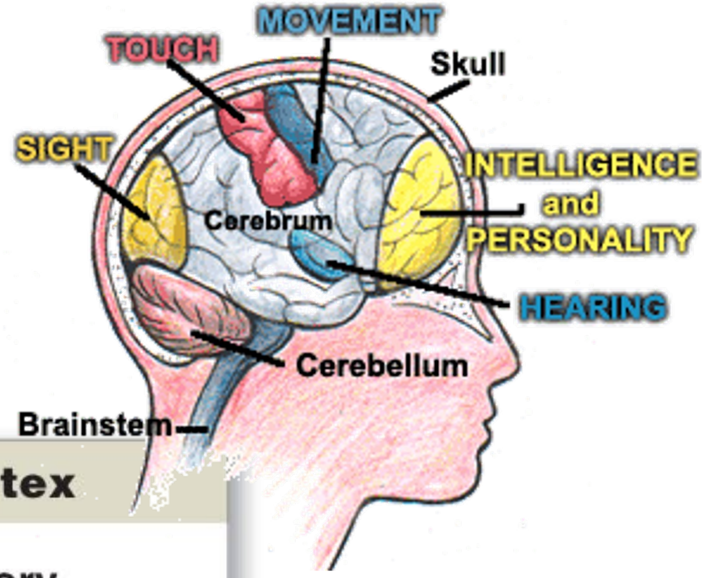
2) Similarity measure between images

- If images of different modality (e.g., MRI and CT) →
- Cross-correlation between patches around corresponding voxels
- Mutual information between intensities at corresponding voxels

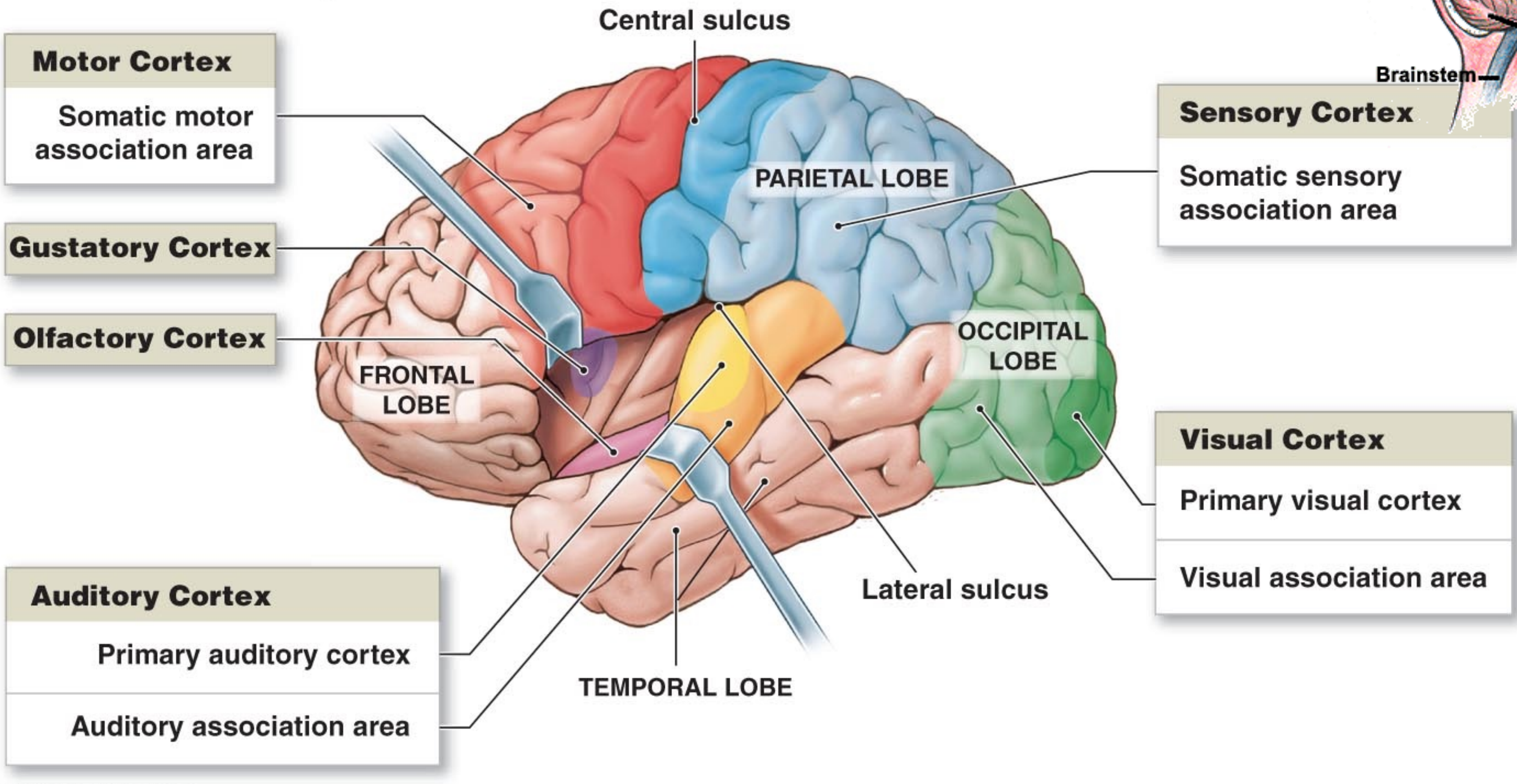


Brain Structure

- Commonality across population

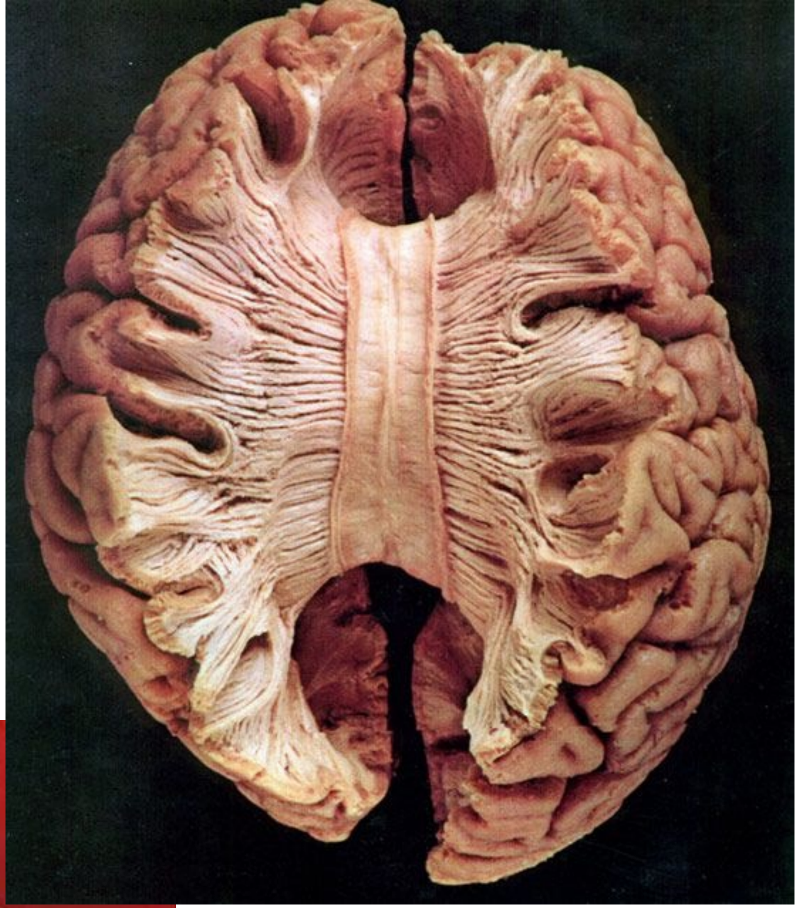


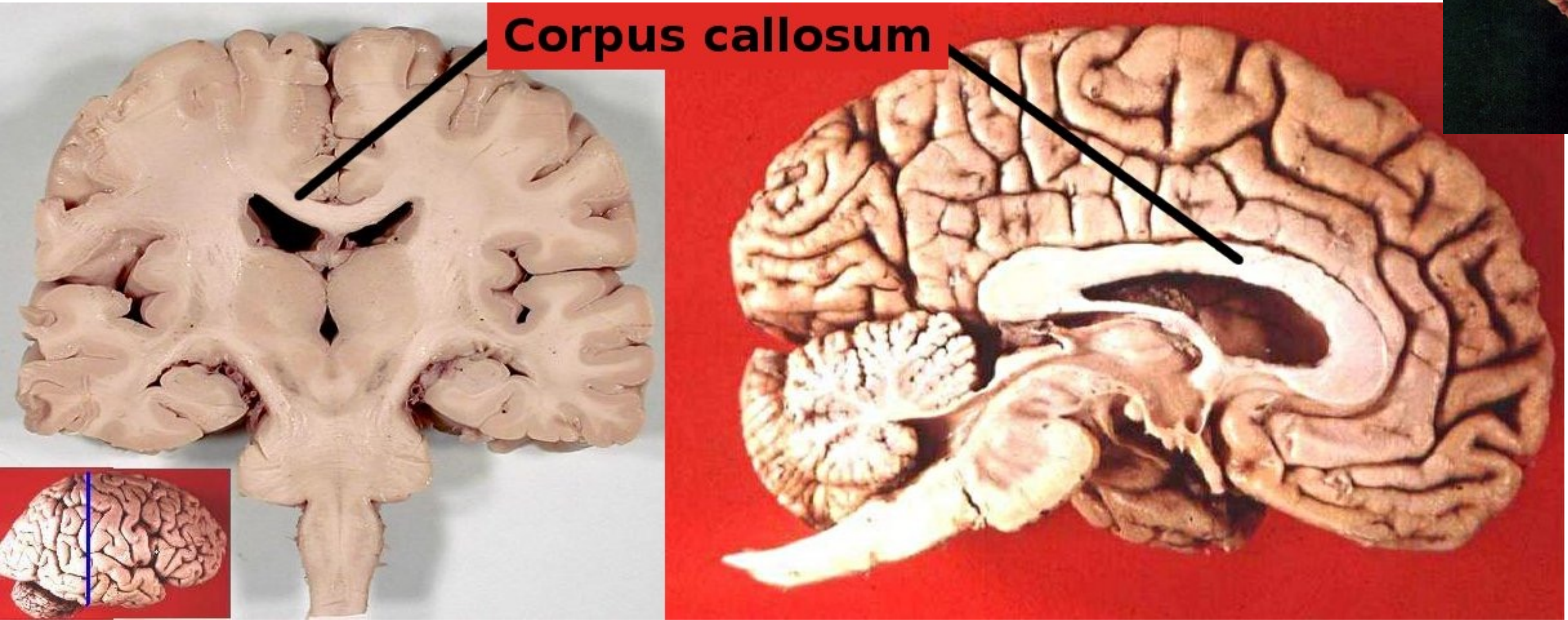
The motor and sensory cortexes and the association areas for each



Brain Structure

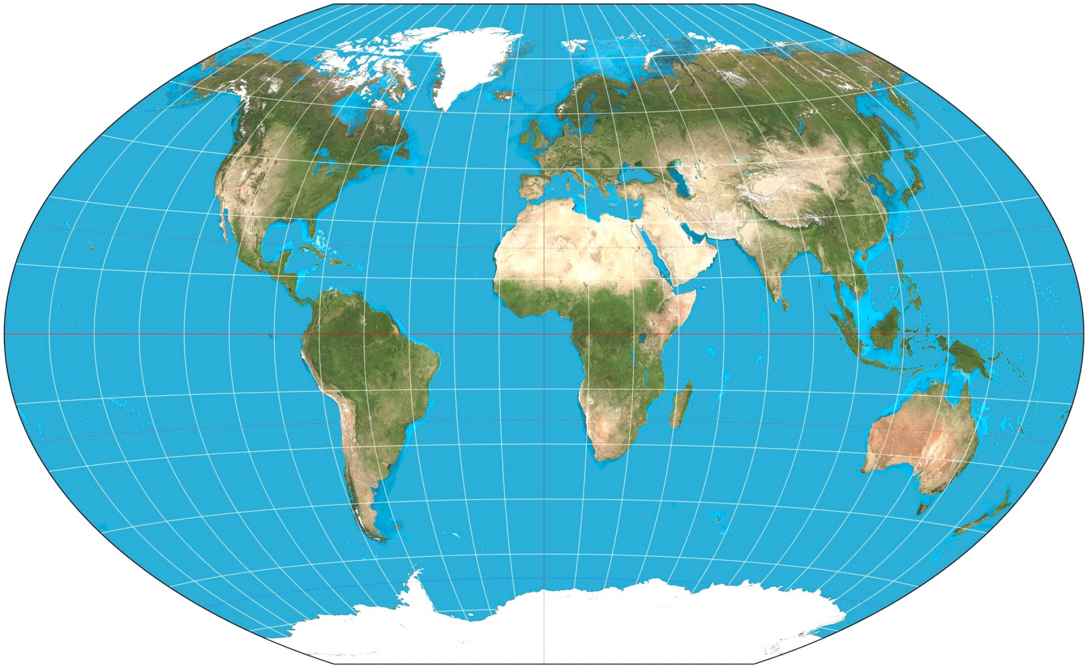
- Commonality across population

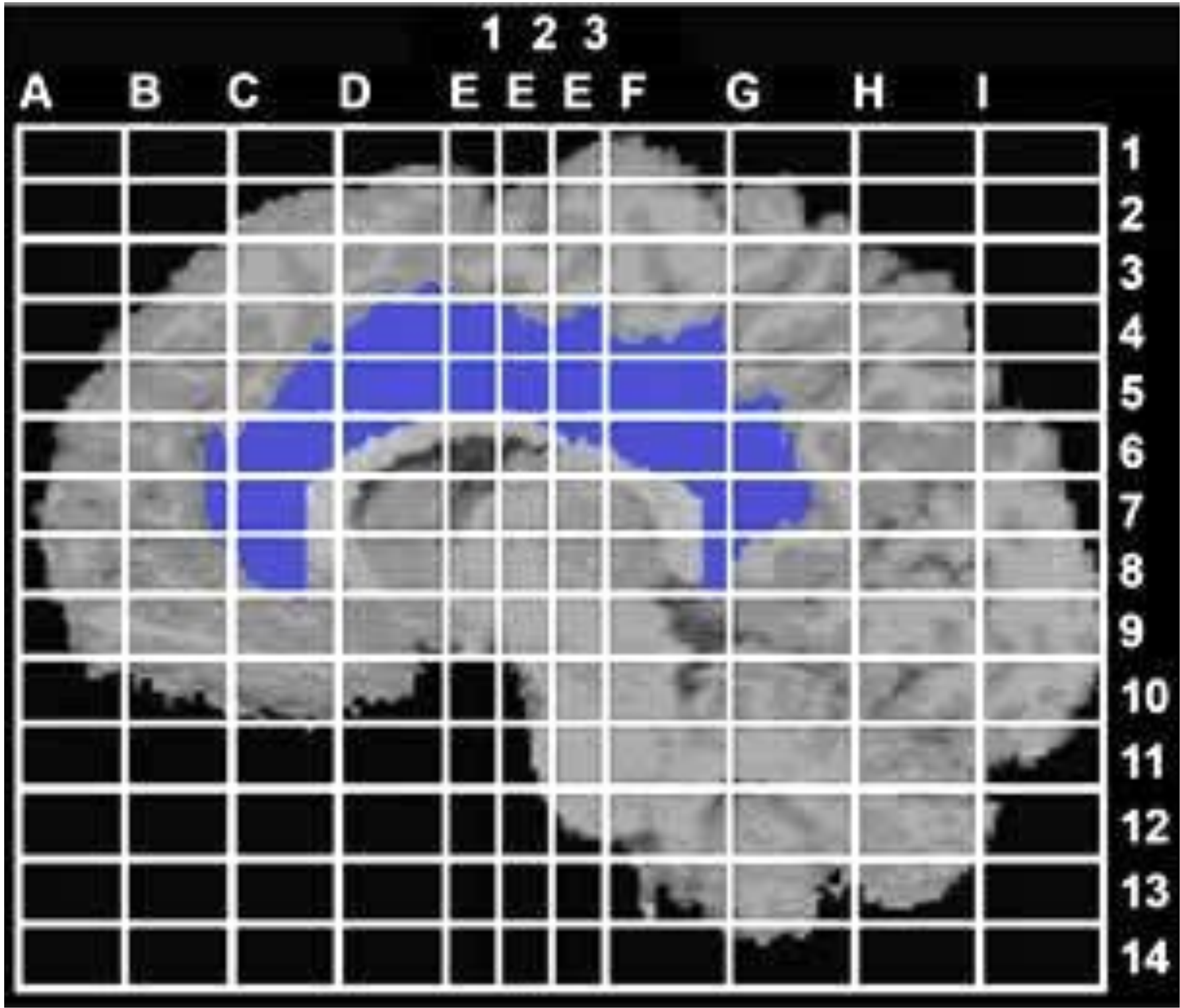




Brain Structure

- Brain Atlas: 3D coordinate system for (typical/average) human brain
  - Used to map location of brain structures independent from individual differences in size and shape
  - Creating a standardized grid for neurosurgery





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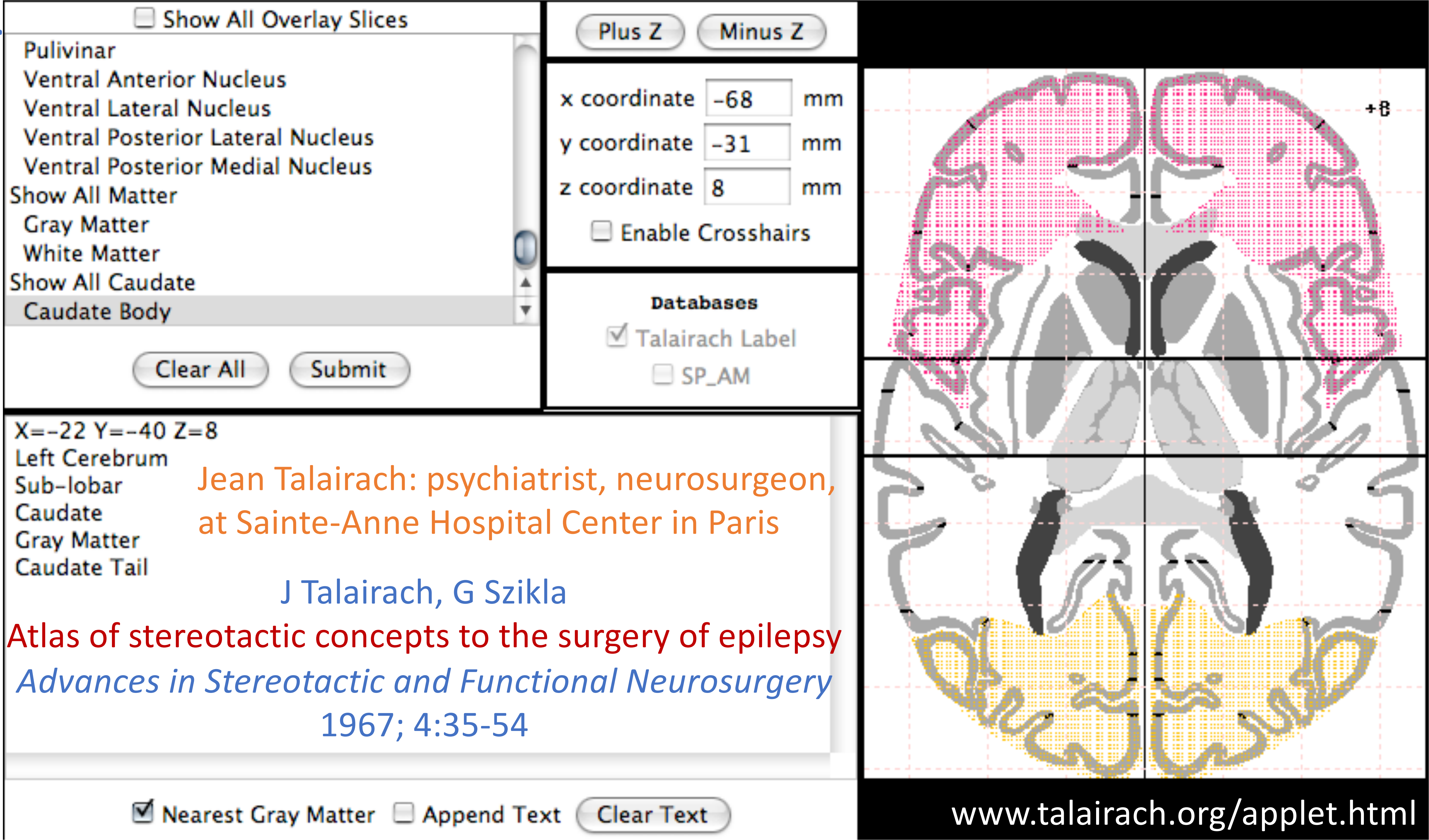


Jean Talairach: psychiatrist, neurosurgeon, at Sainte-Anne Hospital Center in Paris

J Talairach, G Szikla

Atlas of stereotactic concepts to the surgery of epilepsy

Advances in Stereotactic and Functional Neurosurgery

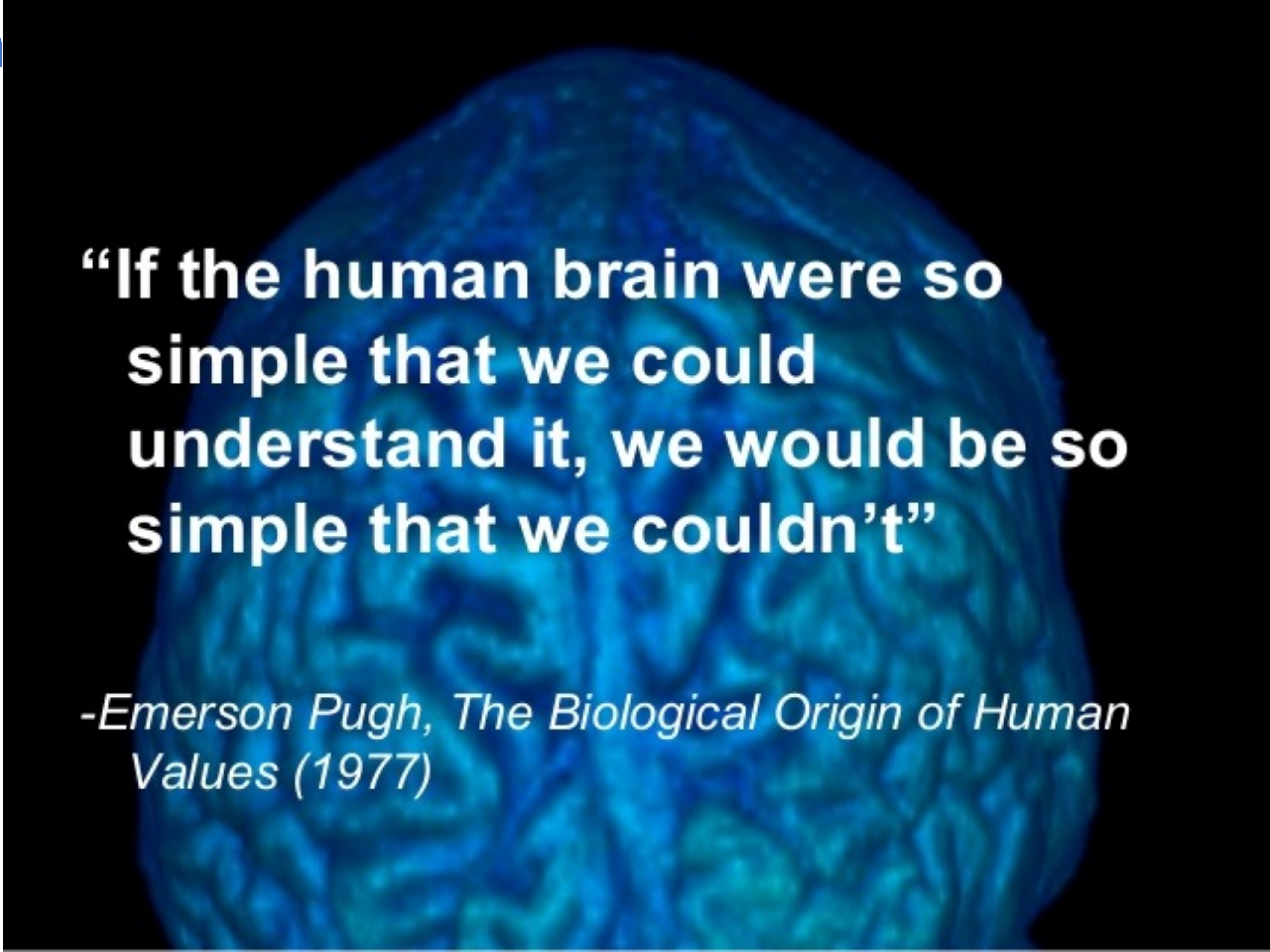
1967; 4:35-54

www.talairach.org/applet.html

Image Registration

- How complex is the human brain ?
- Chat-GPT 5 (2025) has how many parameters ?
  - Around 2-5 trillion parameters
- alleninstitute.org/news/why-is-the-human-brain-so-difficult-to-understand-we-asked-4-neuroscientists
- Number of neurons = 86 billion
- Number of synapses (connection between neurons) = 1 quadrillion = 1000 trillion
- Each synapse contains different molecular switches → 1 synapse isn't equivalent to 1 transistor, but more like 1000 transistors

Image Registra



Emerson Pugh: Engineer at IBM, President of IEEE in 1989.

Image Registration

- Application: *Nature* 2015. Construction of brain atlases based on a multi-center MRI dataset of 2020 Chinese adults.
  Average geometry and appearance → Atlas (for varying age groups)

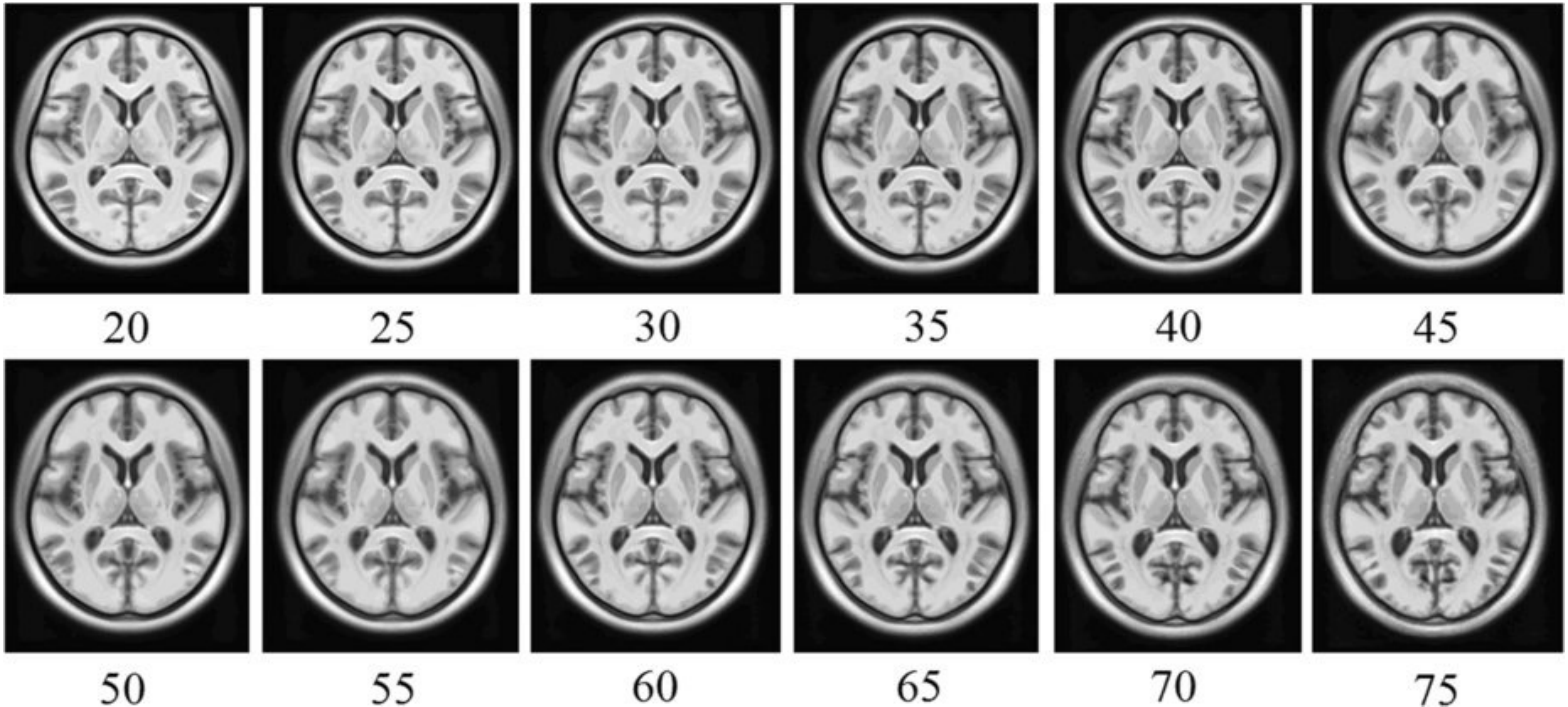


Image Registration

- "Group registration" to find "group mean"
- Method similar to that used for statistical shape analysis
  - Align all to mean, update mean, repeat

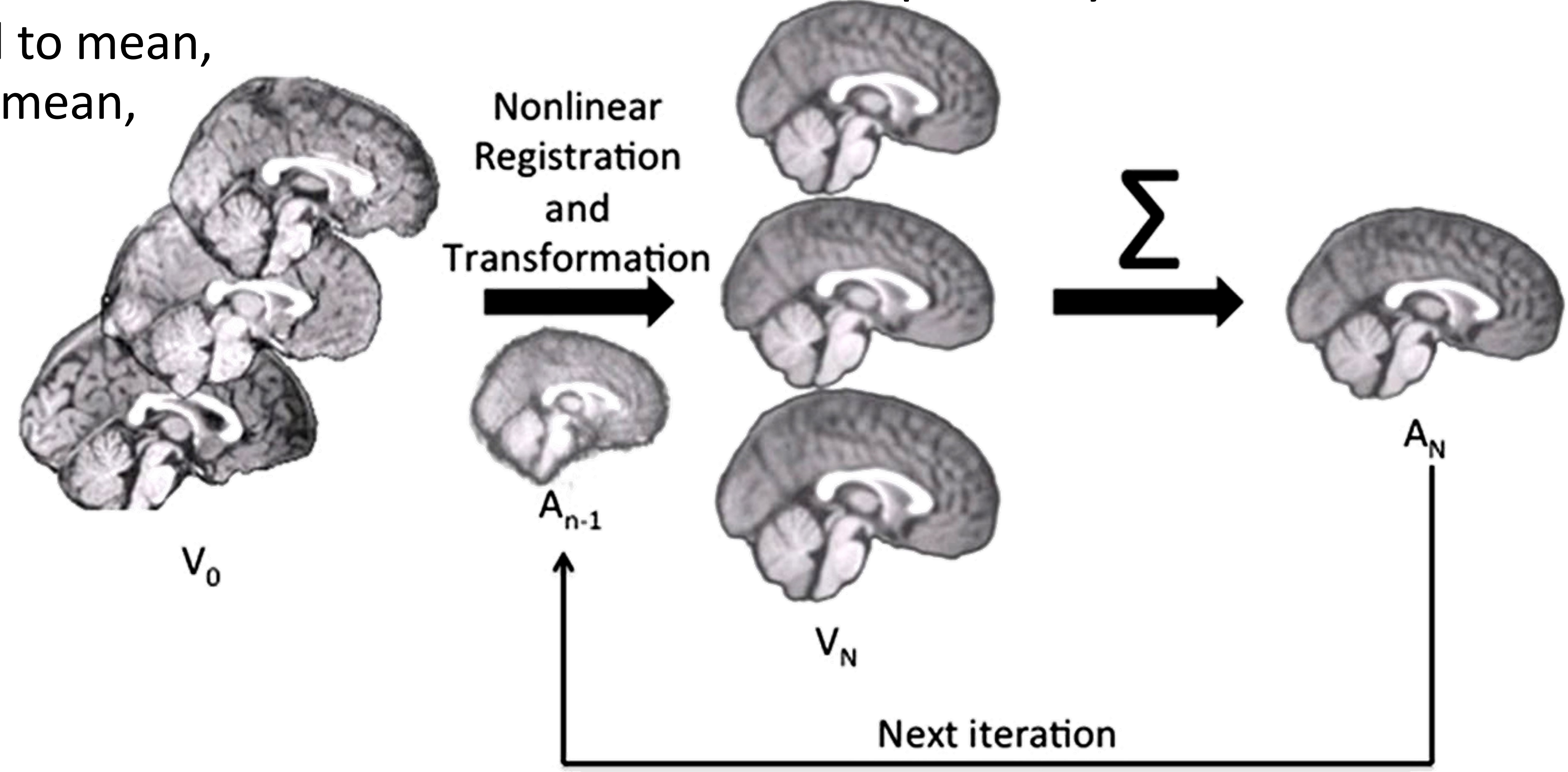


Image Registration

- Let $x \in R^3$ be a (pixel) location in 3D domain Ω
- Given
  - Function R(x): “reference” image
  - Function T(x): “moving” image
  - Example brain-MRI volumes →
- Goal
  - Align image T to image R

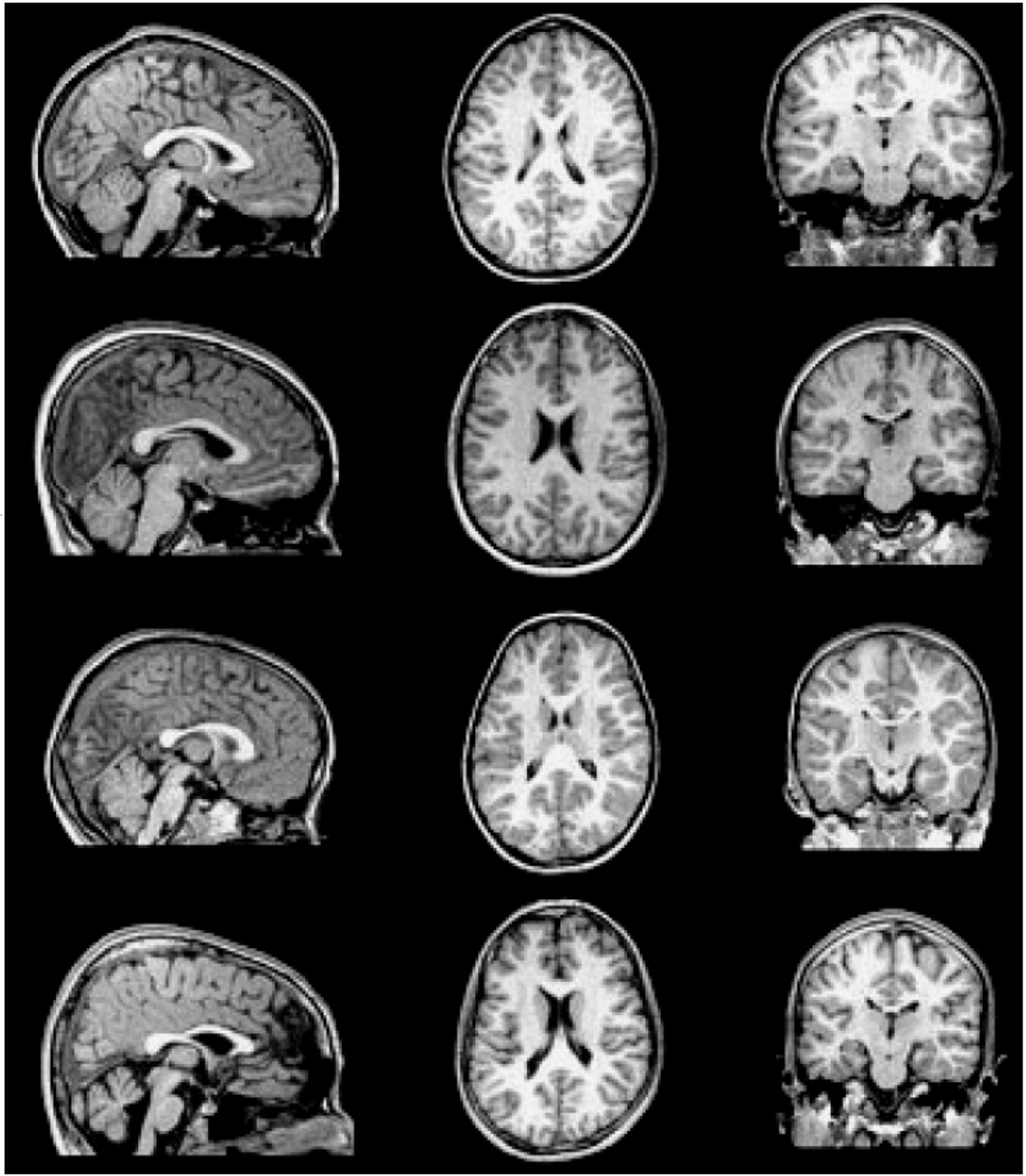


Image Registration

- Formulation
- R(.) and T(.) are discretized images, N pixels with locations $x_i \in R^3$
- Let $\phi(x)$ be a spatial transformation function
- Special case: $\phi(x) := Ax + b$
  - Affine/"linear" transformation/registration, $b \in R^3$, $A \in R^{3x3}$ (invertible)
- Special case for dissimilarity: sum squared differences
- Objective function: minimize over A,b

$$J(A,b) := \sum_{i=1}^{N} \Big(T(\phi(x_i; A, b)) - R(x_i) \Big)^2$$

Image Registration

$$J(A,b) := \sum_{i=1}^{N} \Big(T(\phi(x_i; A, b)) - R(x_i) \Big)^2$$

- Gradient descent w.r.t. translation 'b'
- Using chain rule, partial derivative of objective function:

$$\frac{\partial J(A,b)}{\partial b} := \sum_{i=1}^{N} 2\Big(T(\phi(x_i; A, b)) - R(x_i) \Big) \frac{\partial T(\phi(x_i; A, b))}{\partial \phi(x_i; A, b)} \frac{\partial \phi(x_i; A, b)}{\partial b}$$

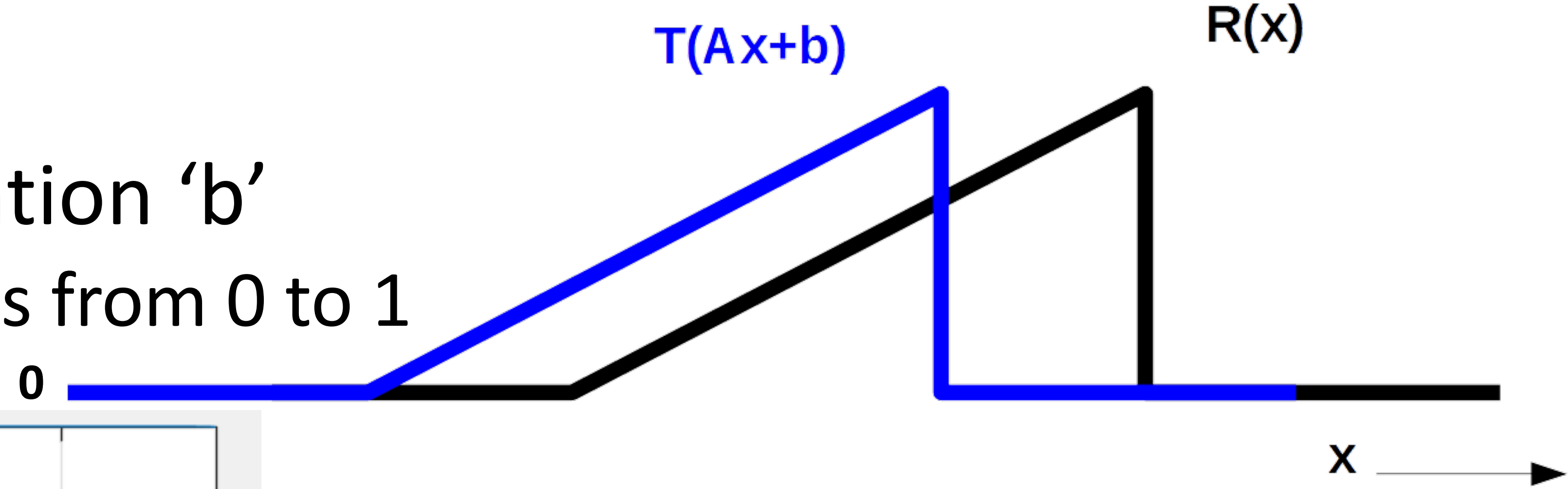
$$= \sum_{i=1}^{N} 2\Big(T(\phi(x_i; A, b)) - R(x_i) \Big) \frac{\partial T(\phi(x_i; A, b))}{\partial \phi(x_i; A, b)}$$

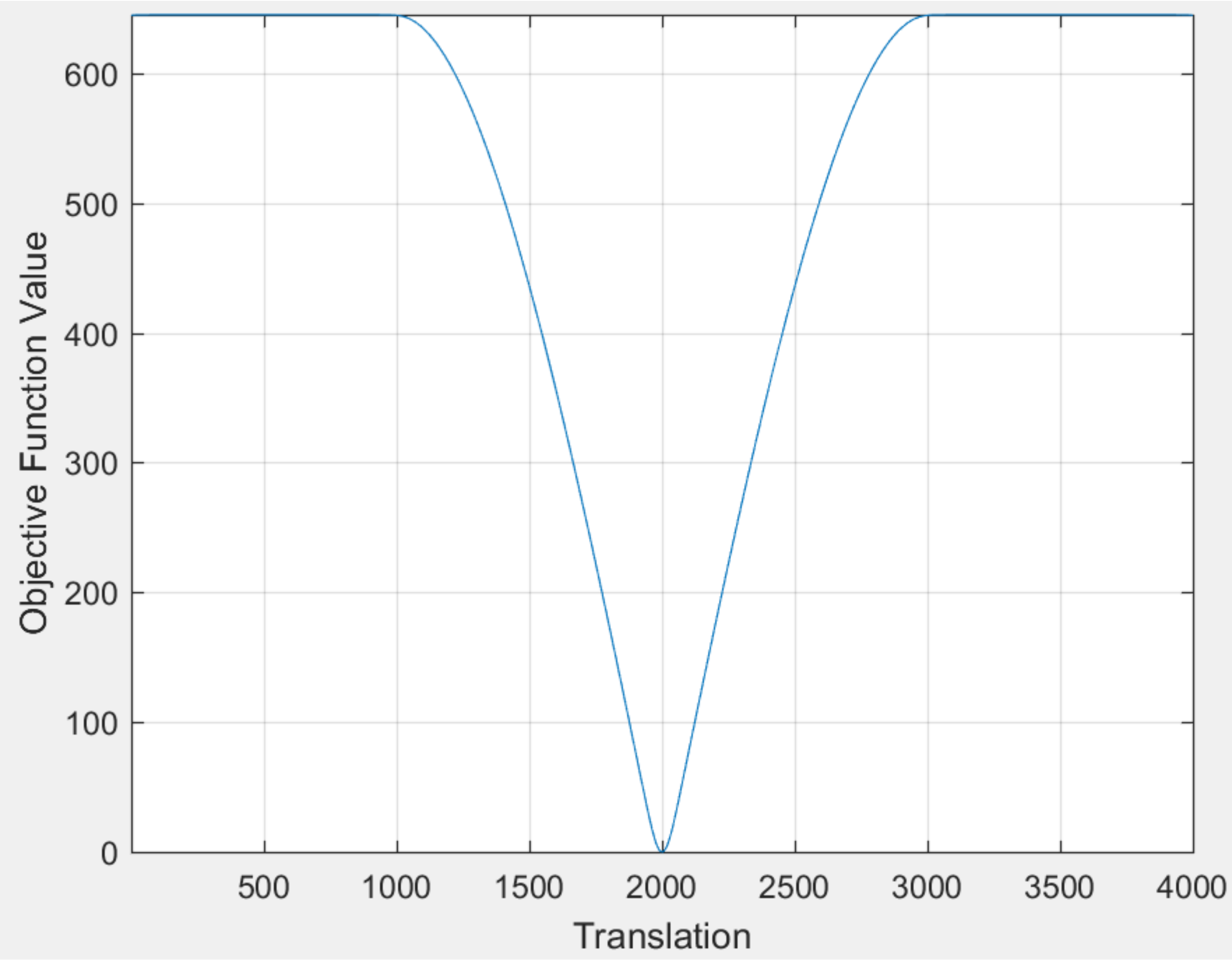
$$\text{because } \frac{\partial}{\partial b}\phi(x) = \frac{\partial}{\partial b}(Ax + b) = \mathbf{I}_{3\times 3}$$

- Spatial derivative of T(.) = row vector (Jacobian of real-valued function)
- Gradient = transpose of derivative (same size as 'b')
    - In this case, this "gradient" is same as well-known "image gradient"

Image Registration

- Gradient descent w.r.t. translation 'b'
  - e.g., ramp functions with values from 0 to 1

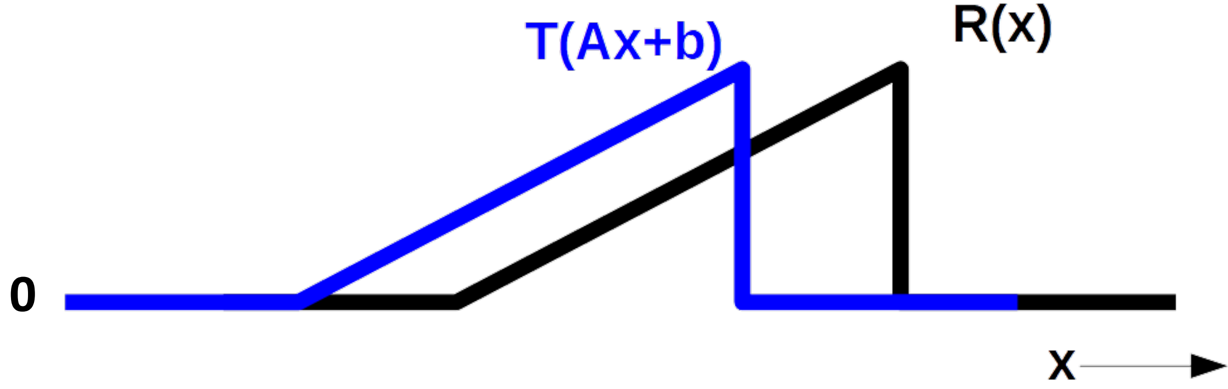




$$J(A,b) := \sum_{i=1}^{N} \Big(T(\phi(x_i; A, b)) - R(x_i) \Big)^2$$

Image Registration

- Gradient descent w.r.t. translation 'b'
  - **For this example:**
  - Φ(x) := Ax + b



$$b \leftarrow b - \tau \frac{\partial J(A,b)}{\partial b}^{\top}$$

$$J(A,b) := \sum_{i=1}^{N} \Big(T(\phi(x_i; A, b)) - R(x_i) \Big)^2$$

$$\sum_{i=1}^{N} 2\Big(T(\phi(x_i; A, b)) - R(x_i) \Big) \frac{\partial T(\phi(x_i; A, b))}{\partial \phi(x_i; A, b)}$$

- 2nd term = spatial derivative of T → "always" pointing right (when T non-zero)
- 1st term = positive (at "all" 'x' where 2nd term > 0)
- Gradient descent reduces 'b', i.e., pushes 'b' left
  - Shifting coordinate axis to left "moves" image T to right
  - Example: let A=Identity; b^{old} = 0; b^{new} = -1.
    Then, intensity in $T^{old}(x)$ at location x=0 now appears at location x=1 in $T^{new}(x)=T^{old}(x-1)$

Image Registration

- Aligning ramps (w.r.t. translation)
  - Area under magenta curve > 0

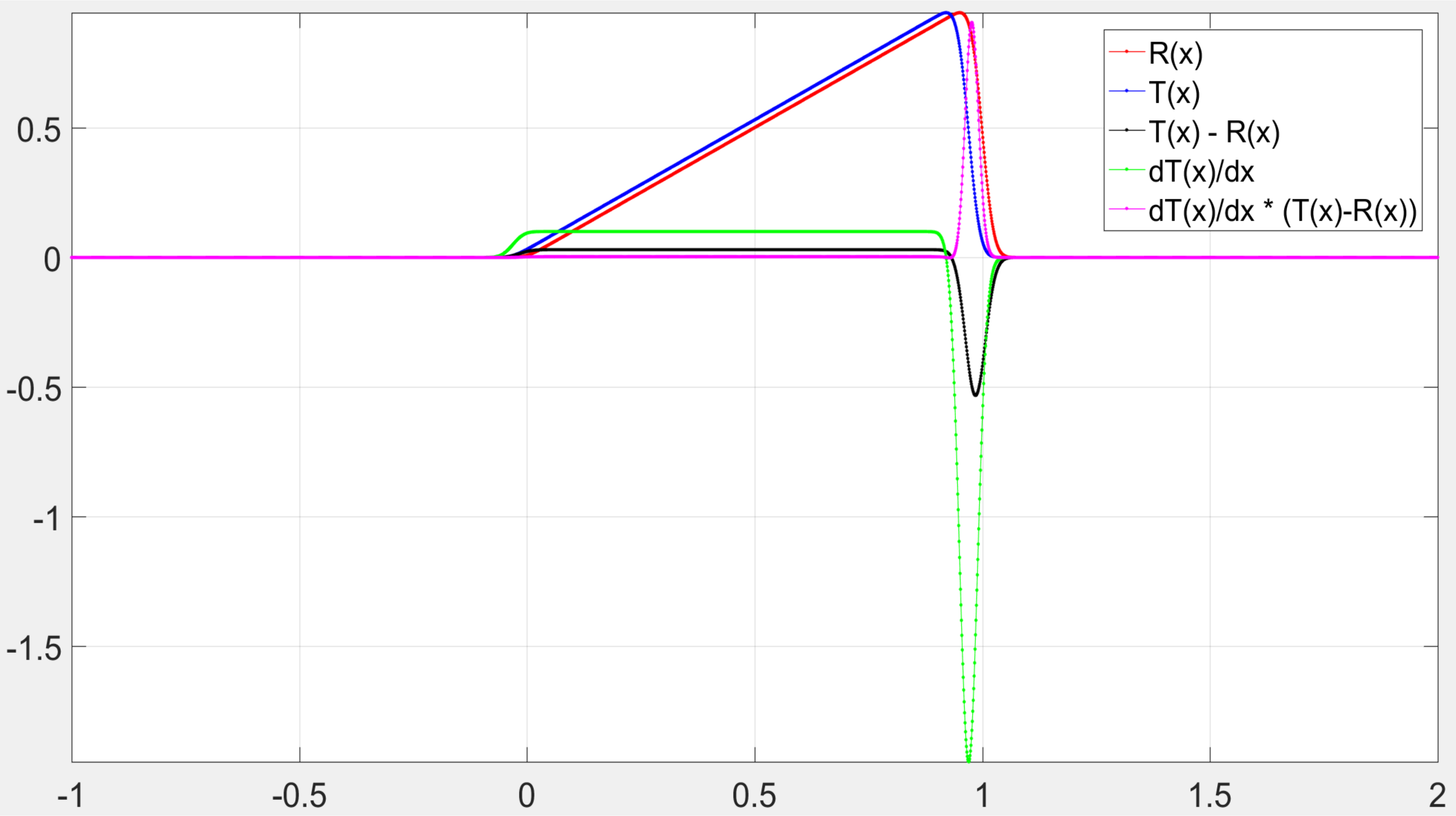


Image Registration

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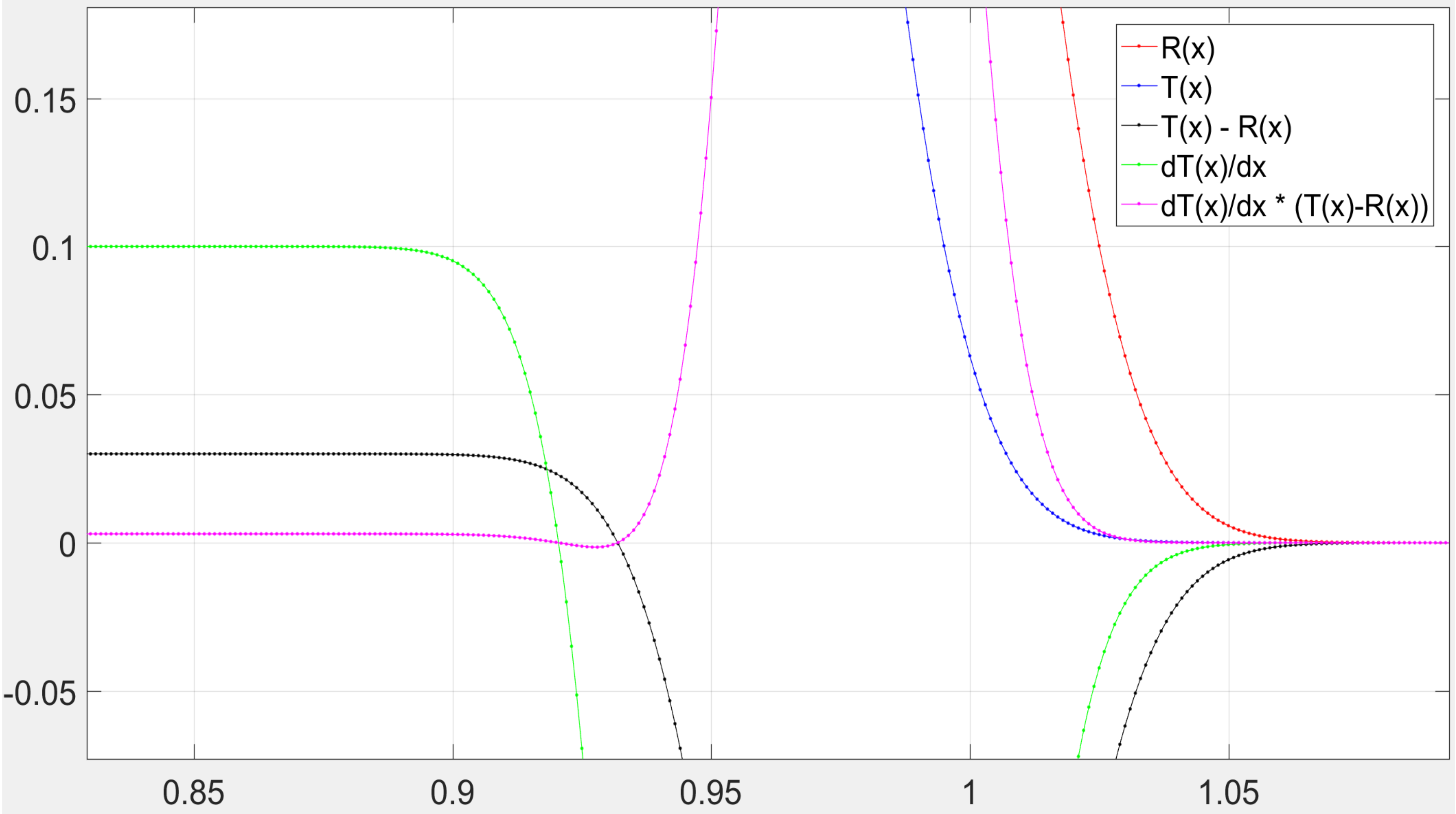

Image Registration

- Aligning ramps (w.r.t. translation)
  - Area under magenta curve **< 0**

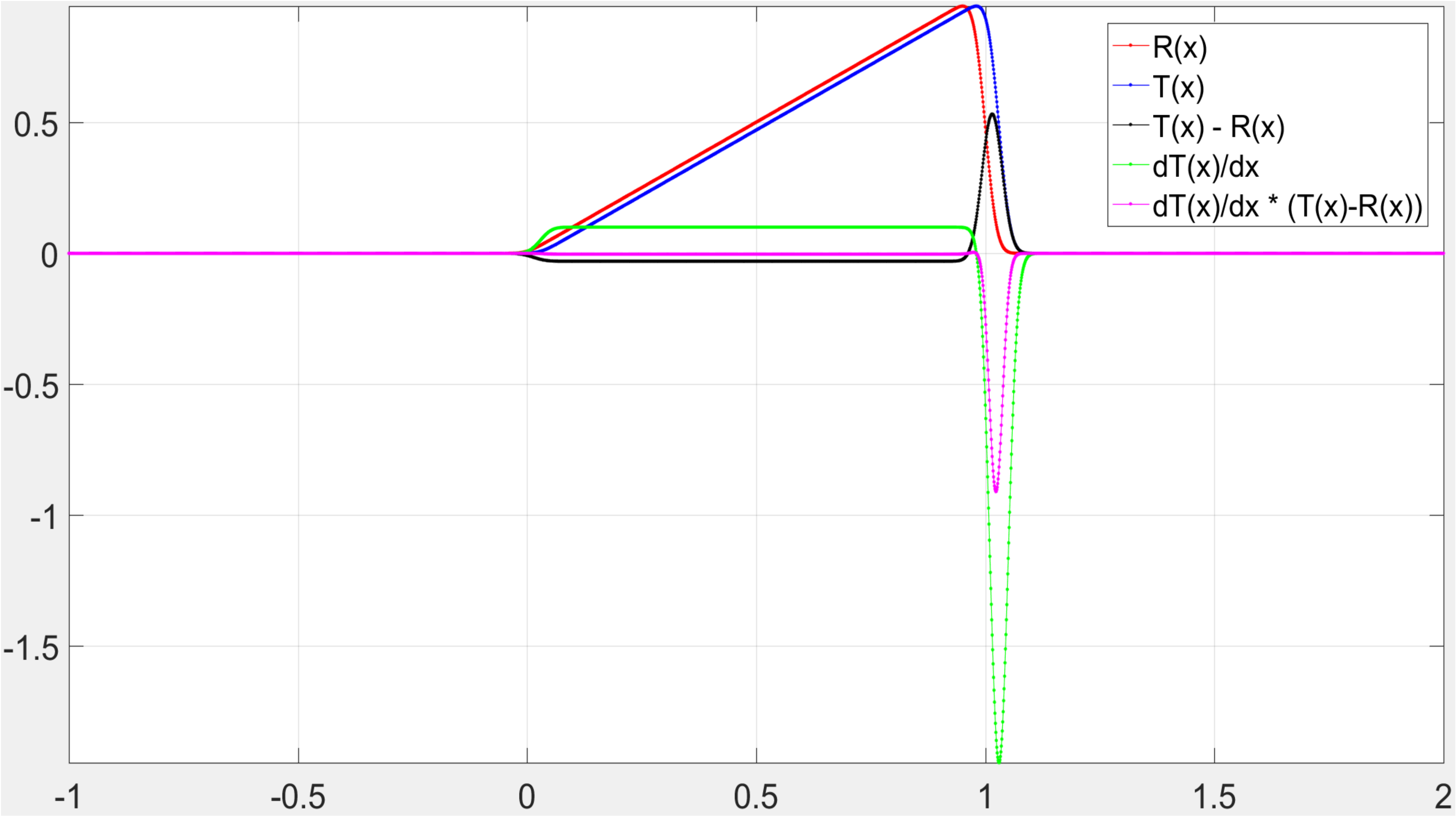

Image Registration

- Aligning ramps (w.r.t. translation)
  - Area under magenta curve **< 0**

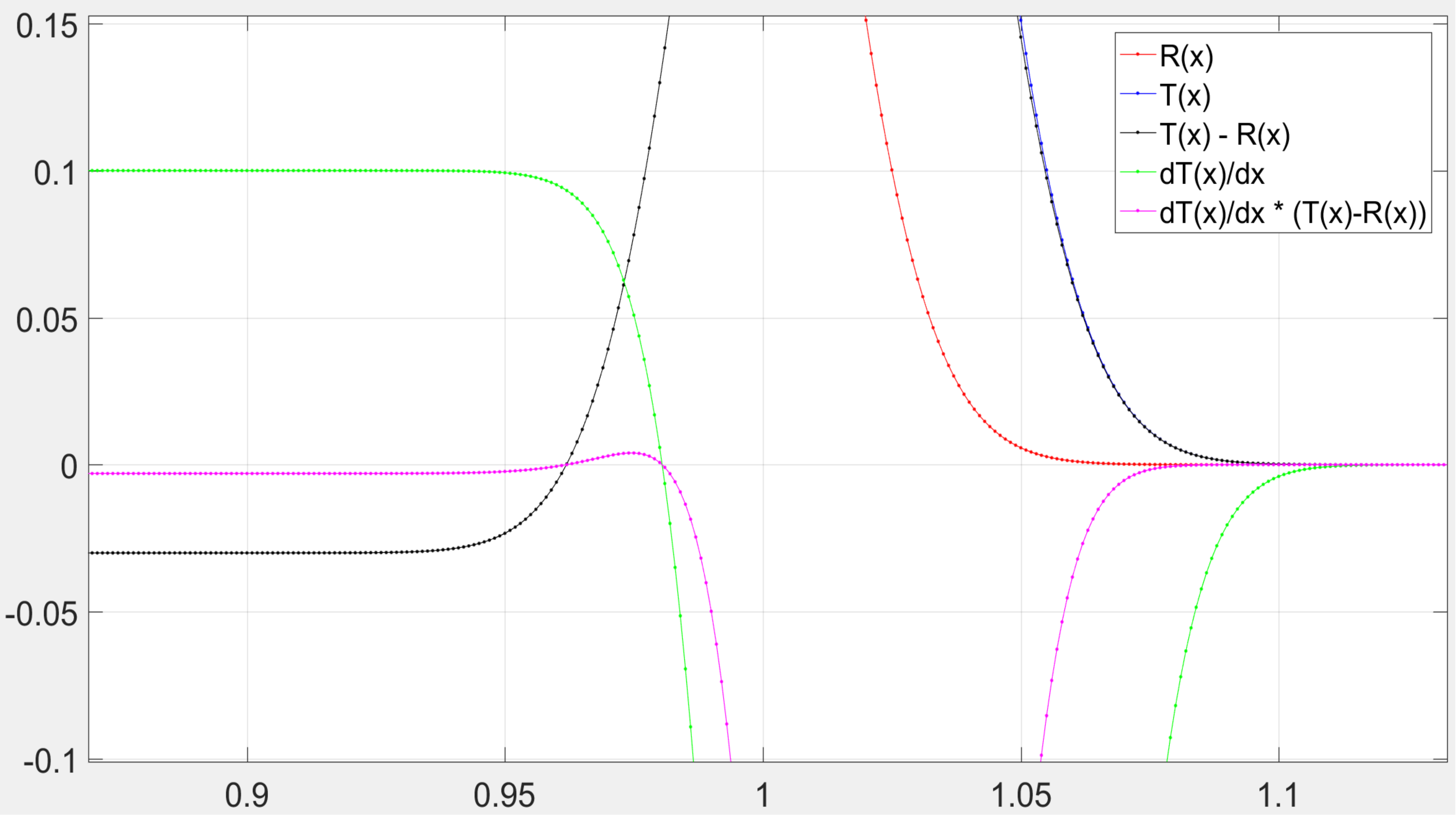

Image Registration

- Align two Gaussians of same height (w.r.t. translation)
  - Area under magenta curve > 0

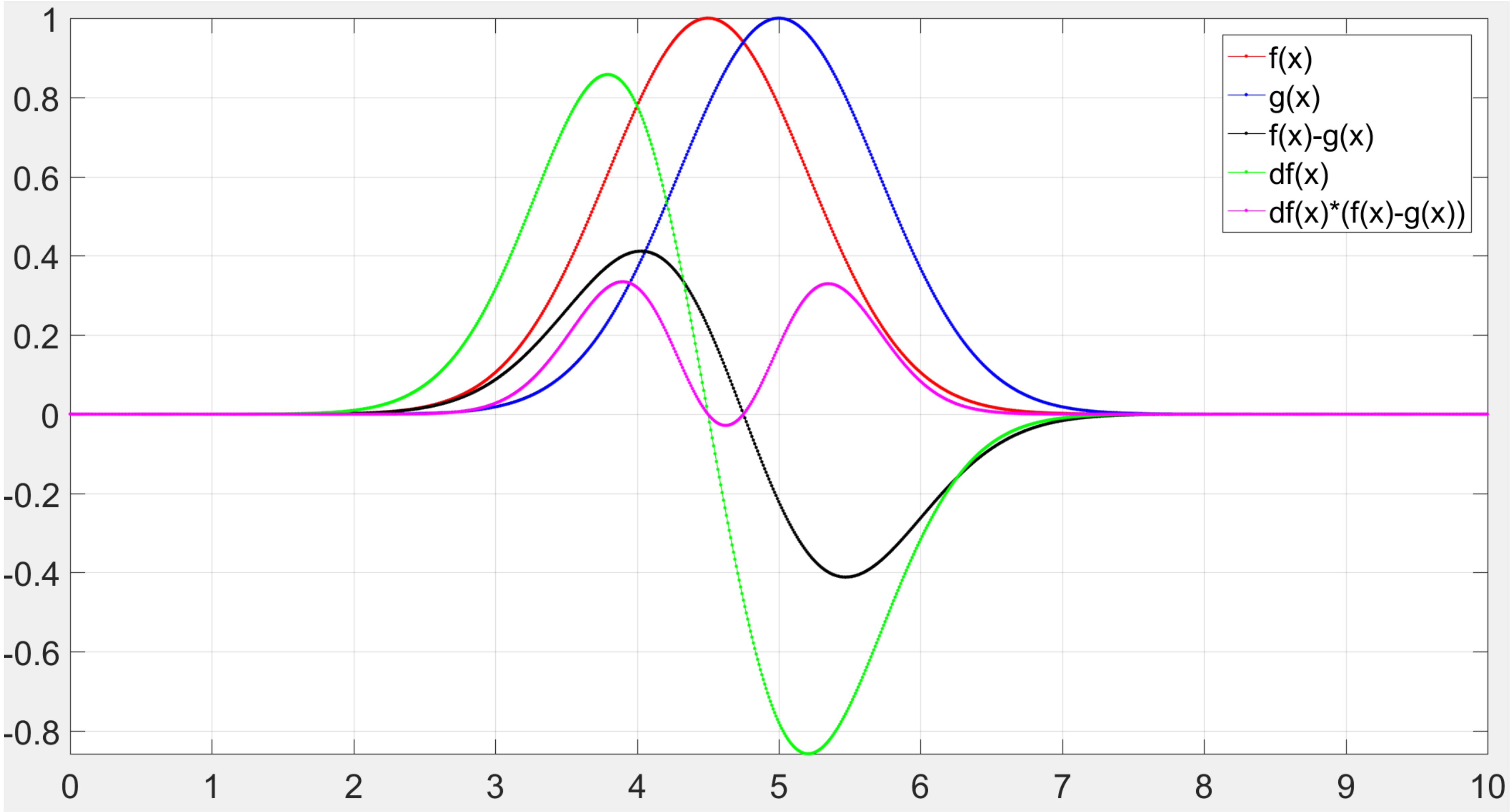

Image Registration

- Align two Gaussians of same height (w.r.t. translation)
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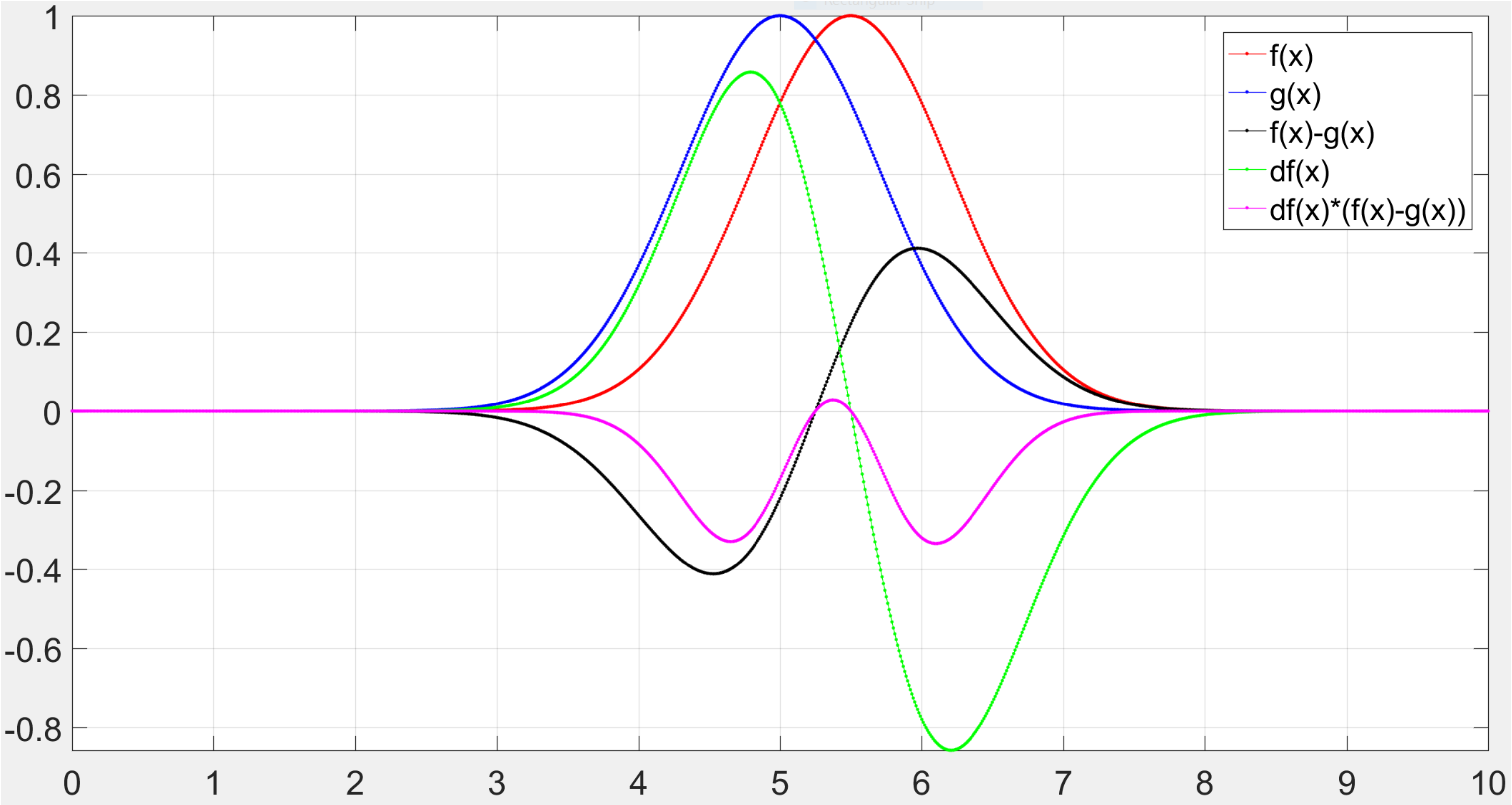


Image Registration

- Align two Gaussians of **different** height (w.r.t. translation)
  - Area under magenta curve > 0

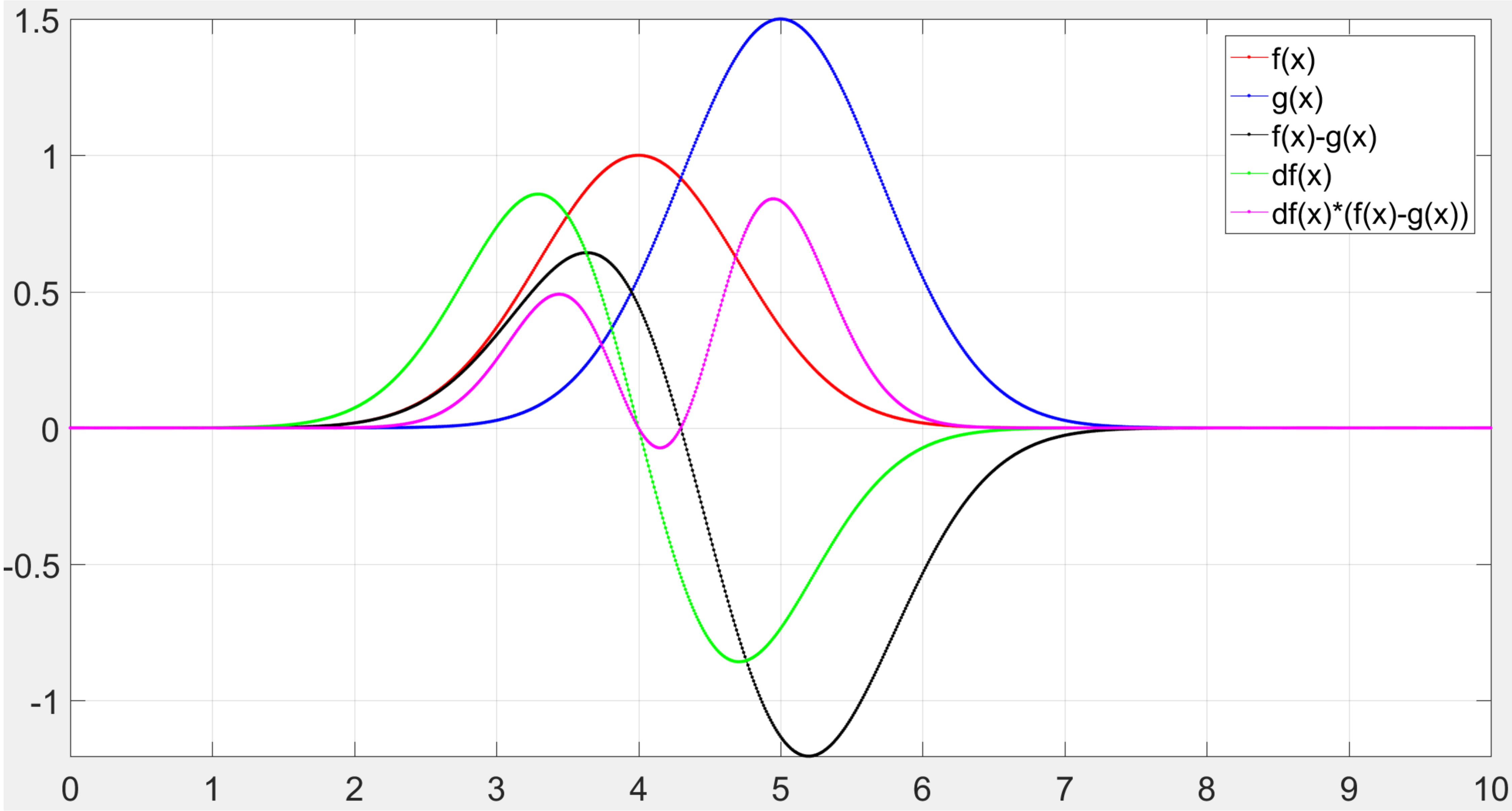


Image Registration

- Align two Gaussians of **different** height (w.r.t. translation)
  - Area under magenta curve > 0

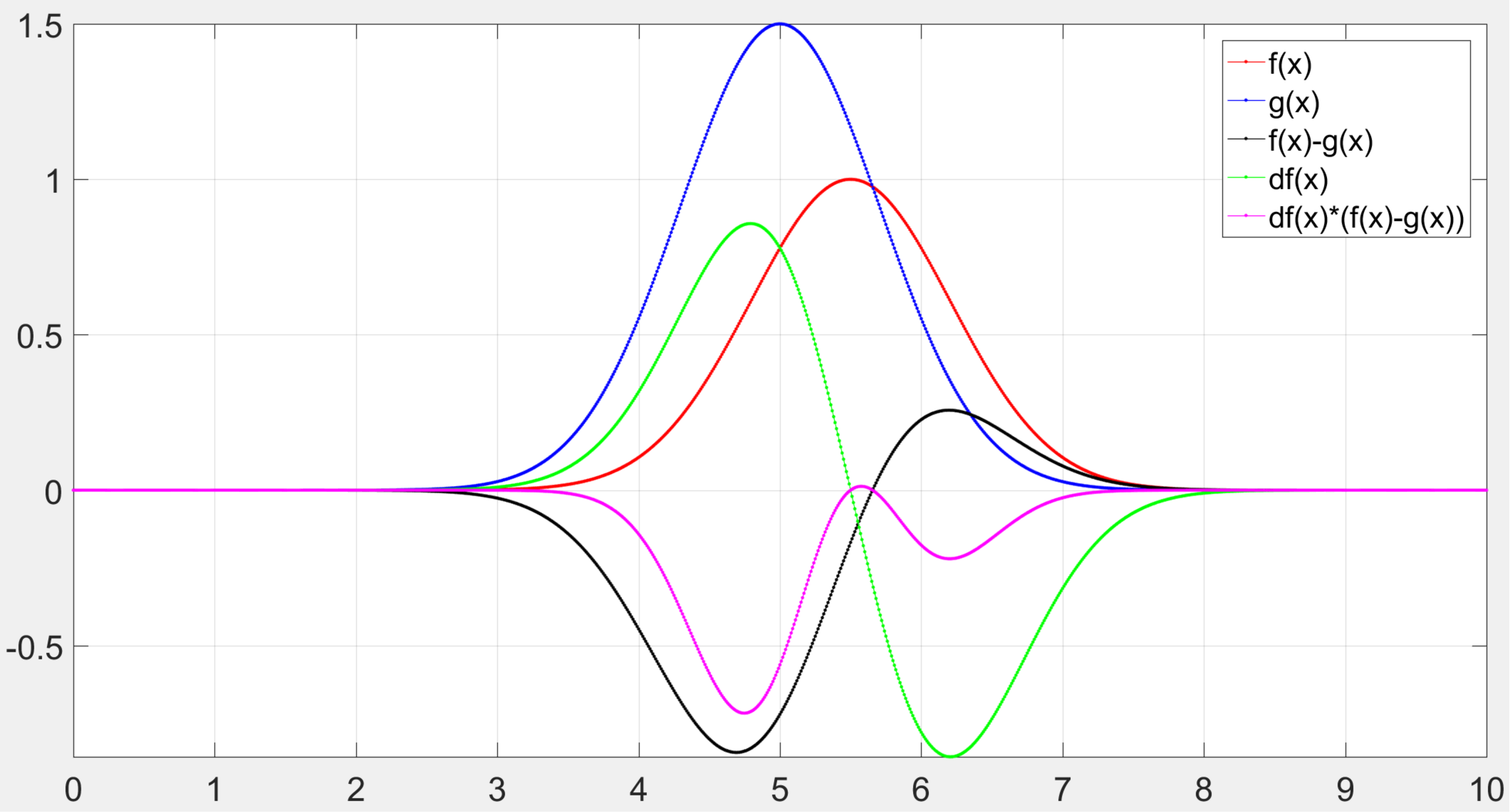


Image Registration

$$J(A,b) := \sum_{i=1}^{N} \Big(T(\phi(x_i; A, b)) - R(x_i) \Big)^2$$

- Gradient descent w.r.t. 'A'
  - Using chain rule, partial derivative of objective function is:

$$\frac{\partial J(A,b)}{\partial A:} := \sum_{i=1}^{N} 2\Big(T(\phi(x_i; A, b)) - R(x_i) \Big) \frac{\partial T(\phi(x_i; A, b))}{\partial \phi(x_i; A, b)} \frac{\partial \phi(x_i; A, b)}{\partial A:}$$

$$= \sum_{i=1}^{N} 2\Big(T(\phi(x_i; A, b)) - R(x_i) \Big) \frac{\partial T(\phi(x_i; A, b))}{\partial \phi(x_i; A, b)} \Big(x_i^\top \otimes \mathbf{I}_{3\times 3} \Big)$$

$$\text{because } \frac{\partial}{\partial A:}\phi(x) = \frac{\partial}{\partial A:}(Ax + b) = x^\top \otimes \mathbf{I}_{3\times 3}$$

  - "A:" = vectorized A by concatenating columns
  - $\otimes$ = Kronecker product →

$$\begin{bmatrix} x_{i1} & 0 & 0 & x_{i2} & 0 & 0 & x_{i3} & 0 & 0 \\ 0 & x_{i1} & 0 & 0 & x_{i2} & 0 & 0 & x_{i3} & 0 \\ 0 & 0 & x_{i1} & 0 & 0 & x_{i2} & 0 & 0 & x_{i3} \end{bmatrix}$$

Image Registration

- Gradient descent w.r.t. 'A'
  - Let image derivative $\frac{\partial T(\phi(x_i;A,b))}{\partial \phi(x_i;A,b)}$ equal the row vector $[\nabla T_{i1}\ \nabla T_{i2}\ \nabla T_{i3}]$
  - Let x_i be $[x_{i1}\ x_{i2}\ x_{i3}]^\top$
  - Then, what is $\frac{\partial T(\phi(x_i;A,b))}{\partial \phi(x_i;A,b)}\left(x_i^\top \otimes \mathbf{I}_{3\times 3}\right)$?

$$[\nabla T_{i1}\ \nabla T_{i2}\ \nabla T_{i3}]\begin{bmatrix} x_{i1} & 0 & 0 & x_{i2} & 0 & 0 & x_{i3} & 0 & 0 \\ 0 & x_{i1} & 0 & 0 & x_{i2} & 0 & 0 & x_{i3} & 0 \\ 0 & 0 & x_{i1} & 0 & 0 & x_{i2} & 0 & 0 & x_{i3} \end{bmatrix}$$

$$=[\ \nabla T_{i1}x_{i1}\ \nabla T_{i2}x_{i1}\ \nabla T_{i3}x_{i1}\ \ \nabla T_{i1}x_{i2}\ \nabla T_{i2}x_{i2}\ \nabla T_{i3}x_{i2}\ \ \nabla T_{i1}x_{i3}\ \nabla T_{i2}x_{i3}\ \nabla T_{i3}x_{i3}\ ]$$

  - Gradient = transpose of derivative (same size as A:)
  - Reshape gradient as a matrix (of same shape as A) $\begin{bmatrix} \nabla T_{i1}x_{i1} & \nabla T_{i1}x_{i2} & \nabla T_{i1}x_{i3} \\ \nabla T_{i2}x_{i1} & \nabla T_{i2}x_{i2} & \nabla T_{i2}x_{i3} \\ \nabla T_{i3}x_{i1} & \nabla T_{i3}x_{i2} & \nabla T_{i3}x_{i3} \end{bmatrix} = \begin{bmatrix} \nabla T_{i1} \\ \nabla T_{i2} \\ \nabla T_{i3} \end{bmatrix} [x_{i1}\ x_{i2}\ x_{i3}]$
    - Outer product of image gradient (column vector) with pixel location (row vector)

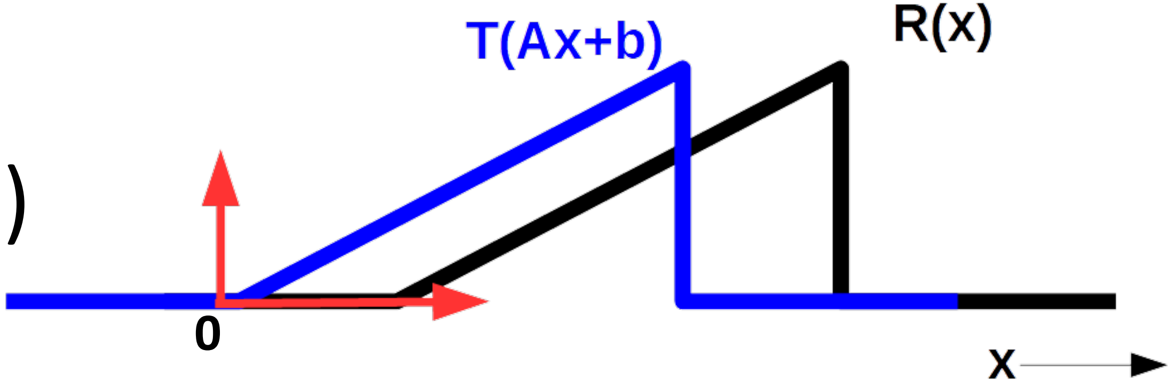
Image Registration

- Gradient descent w.r.t. 'A'

$$A \leftarrow A - \tau \sum_{i=1}^{N} 2\Big(T(\phi(x_i; A, b)) - R(x_i)\Big) \begin{bmatrix} \nabla T_{i1} \\ \nabla T_{i2} \\ \nabla T_{i3} \end{bmatrix} [x_{i1}\ x_{i2}\ x_{i3}]$$

$$= A - \tau \sum_{i=1}^{N} 2\Big(T(\phi(x_i; A, b)) - R(x_i)\Big) \nabla T(\phi(x_i)) x_i^\top$$

Image Registration



- Gradient descent w.r.t. "scaling" 'a' (in 1D)
  - **For this example:**
  - $\Phi(x) := \mathbf{a}x + b$
  - Assume b=0, a>0

$$A - \tau \sum_{i=1}^{N} 2\Big(T(\phi(x_i; A, b)) - R(x_i)\Big) \nabla T(\phi(x_i)) x_i^T$$

  - 2nd term = spatial derivative of T → "always" non-negative
  - 1st term = positive (at "all" 'x' where 2nd term > 0)
  - If 'x_i' all positive, then gradient descent reduces 'a', i.e., shrinks x-axis scale
  - Shrinking coordinate axis → stretches image T to right
  - Example: let b=0; $a^{old} = 1$; $a^{new} = 0.5$.
    Then, intensity in $T^{old}(x)$ at location x=1
    now appears at location x=2 in $T^{new}(x)=T^{old}(ax)$

Image Registration

$$A - \tau \sum_{i=1}^{N} 2\Big(T(\phi(x_i; A, b)) - R(x_i)\Big)\nabla T(\phi(x_i))x_i^{\top}$$

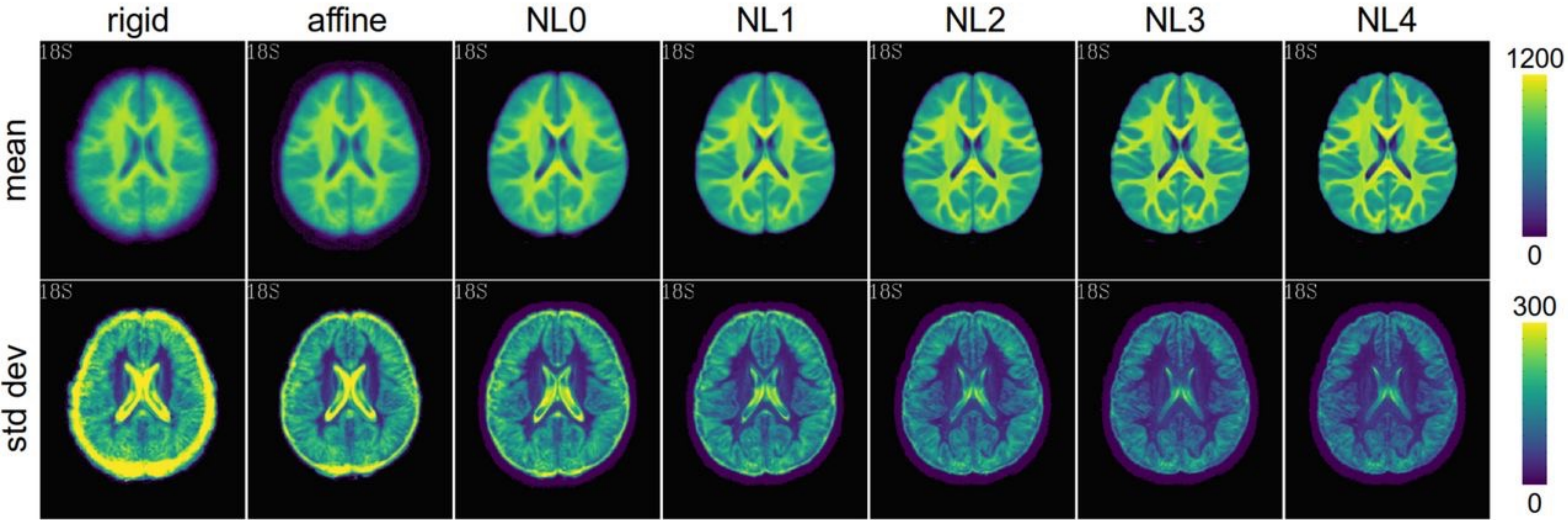
- Gradient descent w.r.t. 'A'
  - Requires computing image gradient ∇T(.) at sub-pixel locations Φ(x), for each descent step (needing multiple interpolations per image-gradient computation)
  - This can become computationally cheaper
  - Substitute y := Φ(x)
  - Replace summation/integration over x by summation/integration over y

$$A \leftarrow A - \tau \sum_{i=1}^{N} 2\Big(T(y_i) - R(\phi^{-1}(y_i; A, b))\Big)\nabla T(y_i)\phi^{-1}(y_i; A, b)^{\top}|A|^{-1}$$

  - $\nabla T(y_i)$ = gradient in image T at pixel location y_i
  - $|A|^{-1}$ = inverse determinant Jacobian of transformation Φ(x) = Ax+b
    - Can be absorbed in selection of step size τ, and ignored

Image Registration

- Atlas construction
  - Effect of choice of deformation model (rigid vs. non-rigid transformation)
    - Rigid → affine → nonLinear0 → … → nonLinear4
      (successive stages during optimization)



A series of five population-specific Indian brain templates and atlases spanning ages 6 to 60 years DOI: 10.1002/hbm.25182

Image Registration

- How to reduce chances of getting stuck
  in local minima,
  especially for images
  with rich textures ?
  - Optimization at
    multiple image scales
    - (Gaussian) scale-space pyramid
    - Similar to annealing
  - First align coarse-scale structures
  - Gradually move to
    alignment of fine-scale features

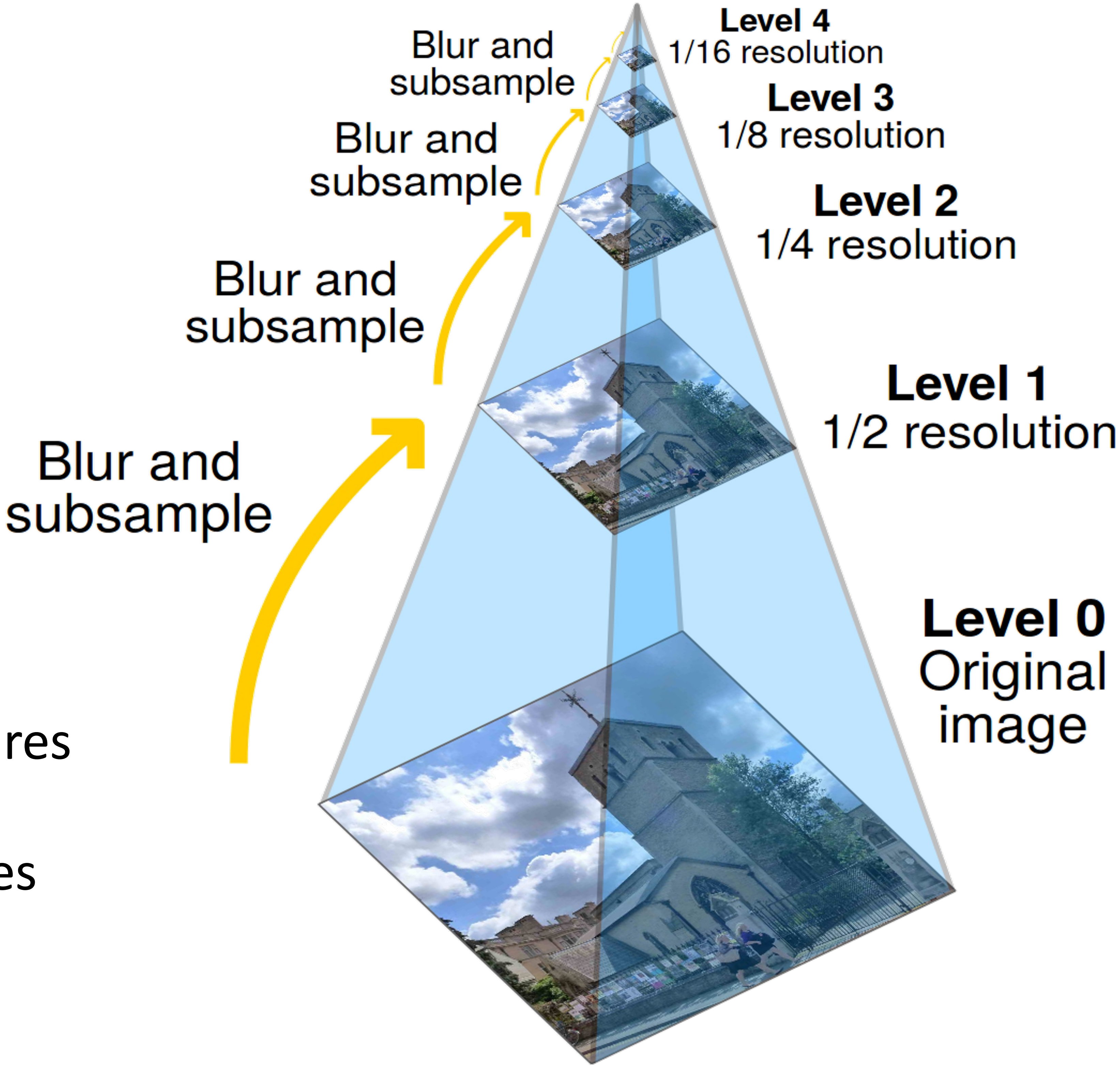
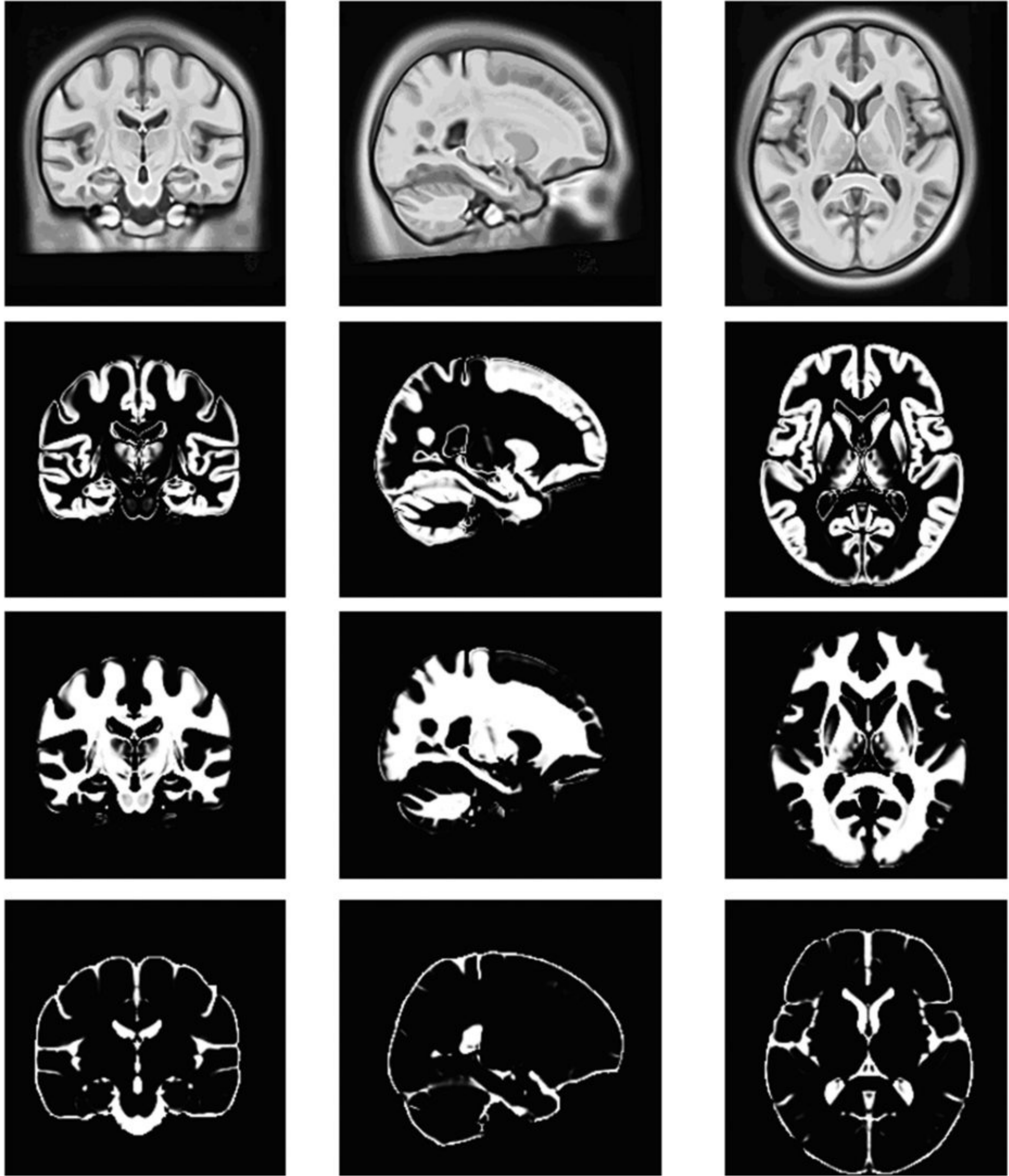


Image Registration

- Applications
- Template (averaged or single image) with
  tissue-probability images



Nature 2015. Construction of brain atlases based on a multi-center MRI dataset of 2020 Chinese adults.

Image Registration

- Applications: segmentation
  - Template with tissue-probability images
  - Left: Template space

Toward Effective Medical Image Analysis Using Hybrid Approaches—Review, Challenges and Applications
DOI:10.3390/info11030155

Right: Subject space

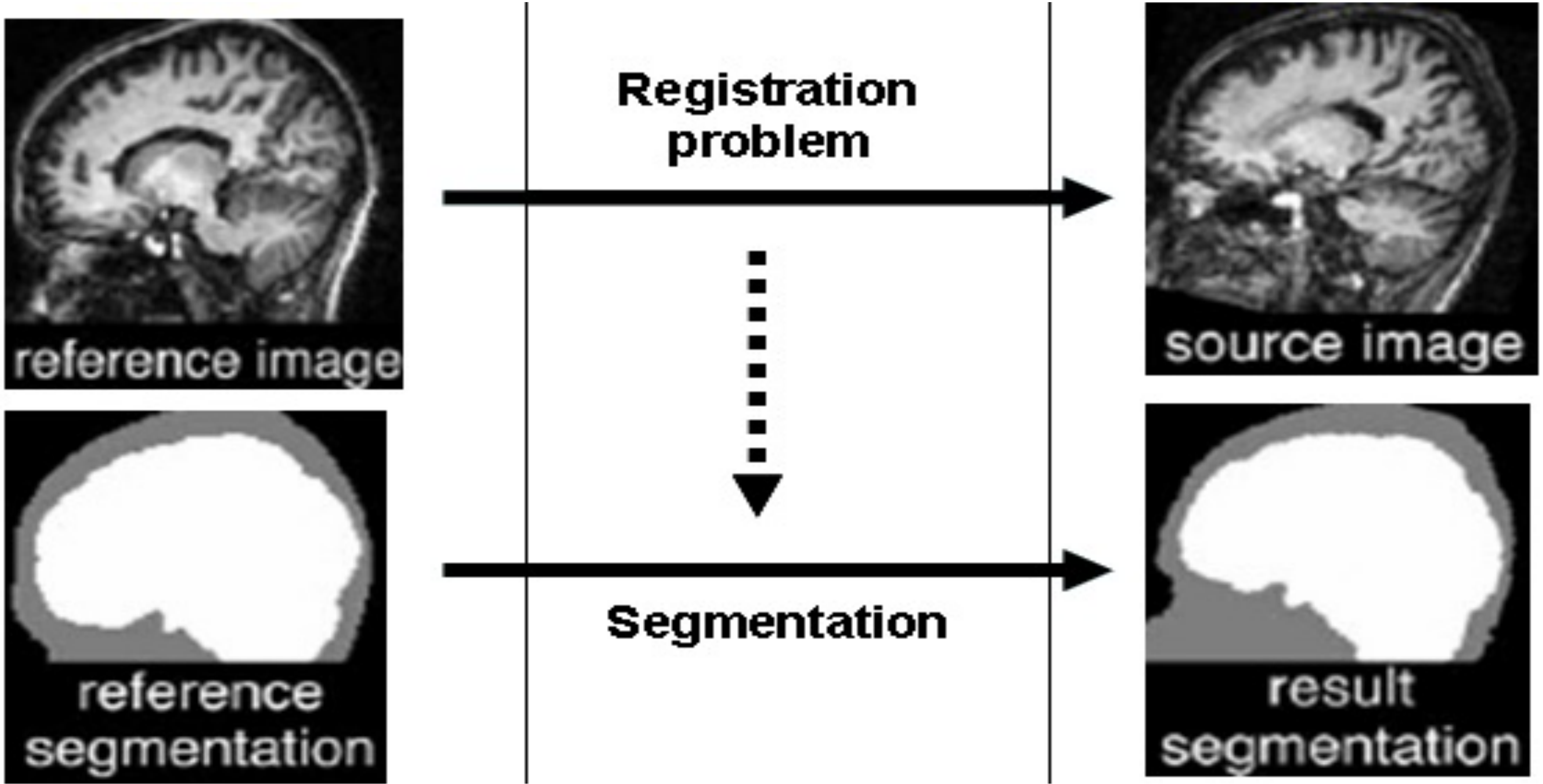


Image Registration

- Applications: segmentation
  - Multiple templates with tissue-probability images

Fully automatic cardiac segmentation from 3D CTA data: a multi-atlas based approach
DOI: 10.1117/12.838370